MIL-STD-1460(MU)
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MILITARY STANDARD

SOLDERING OF ELECTRICAL CONNECTIONS AND PRINTED WIRING ASSEMBLIES, PROCEDURE FOR

FSC 1140

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DEPARTMENT OF DEFENSE

WASHINGTON, D. C. 20301

Soldering of Electrical Connections
and Printed Wiring Assemblies, Procedure for

MIL-STD-1460(MU)

1. This military standard has been approved by the Picatinny Arsenal, Department of the Army and shall be used by that activity. All other military activities are required to employ this standard, where suitable.

2. Recommended corrections, additions, or deletions should be addressed to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-N-S, Dover, New Jersey 07801.

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FOREWORD

The soldering procedures covered by this military standard are to be utilized for the soldering of highly reliable electrical connections in the electrical and electronic equipment of military systems. This military standard outlines the basic soldering procedures which are to be employed. It also graphically illustrates the accept/reject criteria for a variety of soldered connections. It is recognized that soldering connections exist other than those dealt with herein. It is intended that these connections shall be handled under the general provisions of this specification.

The use of methods other than hand soldering is acceptable provided the same quality standards are applied to the finished product and the finished product complies with the requirements specified herein.

This military standard includes some information obtained from MIL-S-45743C(MI), MIL-S-46844A(MI), and NASA publication NPC 200.

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CONTENTS

Paragraph		Page
1.	SCOPE	1
1.1	Scope	1
2.	REFERENCED DOCUMENTS	1
2.1	Government documents	1
2.2	Non-Government documents	2
3.	DEFINITIONS	2
4.	GENERAL REQUIREMENTS	5
4.1	Work area environmental controls	5
4.1.1	Cleanliness	5
4.1.2	Protective packaging	5
4.1.3	Temperature	5
4.1.4	Noise	5
4.1.5	Exhaust	5
4.1.6	Lighting	5
4.1.7	Interior Finish	5
4.2	Materials for soldering connectors and terminals	5
4.2.1	Materials	5
4.2.2	Solder	6
4.2.3	Flux	6
4.2.4	Cleaning solvents	6
4.2.5	Flexible insulation tubing or sleeving	6
4.2.6	Terminals	7
4.2.7	Wire	7
4.2.8	Solderability	7
4.3	Materials for soldering printed wiring	7
4.3.1	Materials	7
4.3.2	Solder	7
4.3.3	Flux	7
4.3.4	Solvents	7
4.3.5	Materials and processes compatibility	8
4.3.6	Printed wiring board solderability	8
4.4	Equipment, tools, and application	8
4.4.1	Tools	8
4.4.2	Brushes and steel wool	8
4.4.3	Pliers	8
4.4.4	Bending tools	8
4.4.5	Clinching tools	8
4.4.6	Insulation strippers	8
4.4.7	Soldering iron tips	8
4.4.8	Soldering irons	9
4.4.9	Soldering iron holder	9
4.4.10	Soldering tools	9
4.4.11	Tools and materials for cleaning	9
4.5	Preparation for soldering	10
4.5.1	Semiconductor soldering precautions	10

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Check the source to verify that this is the current version before use.

CONTENTS (Continued)

Paragraph		Page
4.5.2	Preparation of conductors	11
4.5.3	Preparation of terminals	12
4.5.4	Mechanical connection of conductors to terminals	12
4.5.5	Shielded cable preparation	14
4.6	General soldering technique	15
4.6.1	Securing conductors	15
4.6.2	Heat application	15
4.6.3	Solder application	16
4.6.4	Soldered connections appearance requirements	17
4.6.5	Removal of flux and impurities	18
4.6.6	Connection inspection	18
4.6.7	Reworking unsatisfactory connections	18
4.7	Soldering of printed circuits	19
4.7.1	Solderability	19
4.7.2	Protection during assembly	19
4.7.3	Preassembly quality requirements	19
4.7.4	Eyelets and tubelets	19
4.7.5	Terminal mounting	19
4.7.6	Mounting and soldering of component parts	20
4.7.7	Soldering to gold plated circuits	20
4.7.8	Post soldering inspection of terminals and boards	21
4.7.9	Conformal coating	21
4.7.10	Protective packaging	21
4.8	Training and certification	21
4.8.1	General requirements	21
4.8.2	Records	21
4.8.3	Certification	21
4.8.4	Maintenance of certification status	21
4.8.5	Recertification	21
4.8.6	Revocation of certified status	22
4.8.7	Vision requirements	22
4.8.8	Visual standards	22
4.9	Workmanship	23
5.	DETAIL REQUIREMENTS	23
5.1	Responsibility for inspection	23
5.1.1	General	23
5.2	Inspection	24
5.2.1	Visual	24
5.2.2	Mechanical	24
5.2.3	Surveillance	24
5.3	Acceptance criteria	24
5.3.1	Solder connection	24
5.3.2	Residual flux	24
5.3.3	Excessive solder	24
5.3.4	Cold solder joint	24
5.3.5	Rosin joint	25
5.3.6	Insulation	25
5.3.7	Conductor wire	25
5.3.8	Conductor wire and component lead tension	25
5.3.9	Excessive heat	25
5.3.10	Shielding	25

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Check the source to verify that this is the current version before use.

CONTENTS (Continued)

Paragraph		Page
5.3.11	Multiple terminations	25
5.4	Rejection criteria	25
5.4.1	General criteria	25
5.4.2	Printed wiring criteria	26
5.5	Rejects	27
5.6	Rework	27
5.7	Inspection standards	27
5.8	Qualification and certification	27
5.9	Work areas and environments	27
6	NOTES	27
6.1	Intended use	27
6.2	Ordering data	27
6.3	Approved products	27
6.3.1	Solvents	28
6.3.2	Flux	28
6.4	Soldering methods	28
6.5	Supersession data	28
APPENDIX	ILLUSTRATIONS AND VISUAL STANDARDS	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:47Z
Check the source to verify that this is the current version before use.

FIGURES

Figure		Page
1.	Stripping tools	A-1
2.	Thermal shunts and anti-wicking tools	A-2
3.	Lead cleaning and bending tools	A-3
4.	Stress relief	A-4
5.	Component stress relief	A-5
6.	Insulation trim	A-6
7.	Insulation trim	A-7
8.	Wire tinning	A-8
9.	Lead bending	A-9
10.	Sleeving application	A-10
11.	Turret terminal wire wrap	A-11
12.	Turret terminal wire wrap	A-12
13.	Bifurcated terminal wire wrap	A-13
14.	Bifurcated terminal wire wrap	A-14
15.	Hook terminal wire wrap	A-15
16.	Hook terminal wire wrap	A-16
17.	Solder cup terminal lead placement	A-17
18.	Shield removal for ground termination	A-18
19.	Ungrounded shield termination	A-19
20.	Ungrounded shield termination	A-20
21.	Solder application	A-21
22.	Acceptable joints	A-22
23.	Unacceptable joints	A-23
24.	Component soldering example	A-24
25.	Component soldering example	A-25
26.	Insulated wire soldering example	A-26
27.	Insulated wire soldering example	A-27
28.	Connector cup soldering examples	A-28
29.	Connector cup soldering examples	A-29
30.	Printed circuit soldering examples	A-30
31.	Printed circuit soldering examples	A-31
32.	Using wick to remove solder	A-32
33.	Printed circuit pattern tinning	A-33
34.	Printed circuit pattern tinning	A-34
35.	Terminal swaging preparation	A-35
36.	Swaged terminals	A-36
37.	Examples of component mounting	A-37
38.	Clinched lead examples	A-38
39.	Clinched lead examples	A-39
40.	Component lead termination, circuit on one side	A-40
41.	Component lead termination, circuit on both sides	A-41
42.	Crossing conductive lines	A-42
43.	Clinched lead solder example	A-43
44.	Clinched lead solder example	A-44
45.	Straight lead solder example	A-45
46.	Straight lead solder example	A-46

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1. SCOPE

1.1 Scope. - This standard prescribes general requirements for the soldering of electrical connections and printed wiring assemblies. (See 6.1 and 6.4.)

2. REFERENCED DOCUMENTS

2.1 Government documents. - The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification.

SPECIFICATIONS

Federal

O-E-760	Ethyl Alcohol
O-T-236	Tetrachloroethylene Perchloroethylene, Technical Grade
QQ-S-571	Solder; Tin Alloy, Lead-Tin Alloy; and Lead Alloy
TT-I-735	Isopropyl Alcohol

Military

MIL-T-7003	Trichloroethylene, Stabilized Degreasing
MIL-I-7444	Insulation Sleeving, Electrical, Flexible
MIL-P-13949	Plastic Sheet; Laminated, Copper Clad (for Printed Wiring)
MIL-F-14256	Flux, Soldering, Liquid (Resin Base)
MIL-I-22076	Insulation Tubing, Electrical, Nonrigid, Vinyl, Very Low Temperature Grade
MIL-I-22129	Insulation Tubing, Electrical, Polytetrafluoroethylene Resin, Nonrigid
MIL-I-23053	Insulation Sleeving, Electrical, Heat Shrinkable; General Specification for
MIL-P-46843	Printed Circuit Assemblies Design and Production of
MIL-S-46844A(MI)	Solder, Bath Soldering of Printed Wiring Assemblies, Automatic Machine Type

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MIL-P-55110	Printed Wiring Board

STANDARDS

MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-429	Printed Wiring and Printed Circuit Terms and Definitions

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Non-Government documents. - The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

EIA-STD-319	Solderability of Printed Circuit Boards

Technical society and technical association specifications and standards are generally available for reference from libraries. They are distributed among technical groups and using Federal agencies.

3. DEFINITIONS*

Bifurcated (Split) Terminal	A terminal containing a slot or split in which wires or leads are placed before soldering.
Birdcaging	Separation of individual strands.
Certification	The act of verifying that required training has been completed and specified proficiency has been demonstrated.
Cold Solder Connection	Unsatisfactory connection resulting from nonwetting and exhibiting an abrupt rise of the solder from the surface being soldered. (See also Disturbed Solder Connection which is often referred to as a "cold joint.")
Component	A part; assembly; or combination of parts, subassemblies, or assemblies mounted to perform a design function.

*Definitions relative to printed wiring and circuits if not contained herein may be obtained from MIL-STD-429.

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Component Lead	The wire, solid or stranded, which extends from and serves as a connection to a component.
Conduction Soldering	Method of soldering which employs a soldering iron for transfer of heat to the soldering area.
Disturbed Solder Connection	Unsatisfactory connection resulting from relative motion between the wire and terminal during solidification of the solder.
Epoxy Potting Compound	A nonconductive, organic plastic used for encapsulation of components and/or wires.
Excessive Solder	Unsatisfactory connections wherein the solder obscures the configuration of the connection.
Eyelet	A tubular metal part having both ends headed or rolled over.
Ground Support	Equipment used to prepare, test, and check launch vehicles and spacecraft prior to launch. Also used to launch, operate, and maintain vehicles and spacecraft in space. This includes static firing test equipment and other test equipment for ground tests whether or not operational during flight operations.
Hook Terminal	A terminal formed in a hook shape.
Installation School for Hand Soldering	A school established at a NASA installation to train agency personnel in Category III for NASA work at that installation or equivalent.
Perforated or Pierced Terminal	A terminal containing a hole through which leads or wires are placed before soldering.
Printed Circuit Pad	A terminal area on a printed circuit used for making electrical connections to the conductive pattern.
Resistance Soldering	Method of soldering by passing a current, which heats the soldering area, by contact with two electrodes (not for sensitive semiconductor components).

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Rosin Flux	A noncorrosive, nonconductive, chemically active compound capable of promoting the wetting of metals with solder by a chemical cleaning action.
Rosin Solder Connection	Unsatisfactory connection which is held together by trapped flux resulting in high electrical resistance.
Solder	A nonferrous, fusible, metallic alloy used to join metallic surfaces by adhesion of the liquid solder.
Terminal	A tie point device used for making electrical connections. Five basic styles of terminals are: bifurcated, hook, perforated or pierced, solder cup and turret.
Thermal Shunt	A device with good heat dissipation characteristics used to conduct heat away from an object.
Tinning	The coating of a surface with a uniform layer of solder.
Tubelet	A tubular metal part with both ends formed in a conical flare of approximately 90 degrees included angle.
Turret Terminal	A round post-type grooved stud around which wires or leads are snugly hooked before soldering. It may have either spacing shoulders or grooves for positioning the wires or leads.
Wet Soldering Iron Tip	Heated soldering iron tip covered with a small quantity of molten solder to accelerate transfer of heat connection.
Wetting	Adhesion of a liquid to a solid surface.
Wicking	The flow of molten solder by capillary action.

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4. GENERAL REQUIREMENTS*

4.1 Work area environmental controls. -

4.1.1 Cleanliness. - Work areas and tools shall be maintained in a clean and orderly condition. All dirt, grease, oil, solder splatter, chips, and other contaminating foreign matter shall be promptly removed. Thermal shunts and anti-wicking tools should be cleaned frequently. Component leads and terminals shall be protected by suitable packaging and minimum handling prior to use. Cleaned component leads, terminals, etc., shall not be handled with bare hands. The use of white lint free fabric gloves is recommended. Smoking, eating, or drinking shall not be permitted in the soldering area. (Drinking fountains are permitted if located at least 6 feet from any soldering bench.)

4.1.2 Protective packaging. - The printed circuit boards shall be protected between operations by storing in a protective container made of a material which does not emit gases or chemicals that could be detrimental to the printed circuitry. The use of "ZIP-LIP" bags is suggested; the use of heat sealed plastic bags is prohibited.

4.1.3 Temperature. - The temperature of the room shall be maintained at 75° ± 10°F and the relative humidity shall be 45±15% unless otherwise specified. Pressure differential shall be maintained positive. Fresh air shall be introduced into the room through a system filter while maintaining a slight positive pressure between the soldering area and adjacent areas to prevent infiltration of dust-laden air.

4.1.4 Noise. - Noise, particularly that generated by ventilating equipment, speaker systems, conversation, mobile equipment, etc., shall be kept to a minimal level to avoid distracting and disturbing soldering and inspecting personnel.

4.1.5 Exhaust. - To prevent contamination of parts and to provide for personnel safety, areas for cleaning parts and areas where toxic or volatile vapors are generated shall have an exhaust system to remove the vapors as they are generated.

4.1.6 Lighting. - Lighting fixtures shall provide a minimum of 150 foot-candles of shadowless illumination on working surfaces.

4.1.7 Interior finish. - All floors, walls, ceilings, and painted fixtures shall be finished with material which provides a surface that is easily cleaned and is resistant to dust, grease, chipping, flaking, or powdering.

4.2 Materials for soldering connectors and terminals. -

4.2.1 Materials. - Except when otherwise specified on detailed drawings, the materials utilized in soldering operations shall conform to the requirements of 4.2.2 through 4.2.8.

*All figures referenced in this section are contained in the accompanying Appendix. The figure number and the page number in the Appendix correspond, i.e., Figure 6 will be found on page A-6 of the Appendix, and Figure 35B will be found on page A-35 of the Appendix.

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4.2.2 Solder. - An Sn60 or Sn63 tin-lead solder, Form W, Type R, RMA, or S conforming to QQ-S-571 shall be used; Form B may be used for lead pretinning.

4.2.3 Flux. - Only rosin flux shall be used in all soldering operations. The use of liquid rosin type flux is permissible for applications such as the removal of excess solder from a joint by wicking (see 4.6.7.2) into stranded conductor wire, and for soldering with solid solder. Where dual usage of cored and solid solder is required, core flux and liquid flux must be compatible. Thinners compatible with the flux shall be used. Liquid rosin flux shall conform to MIL-F-14256, Type RMA, except that the copper mirror test (see 3.5 of MIL-F-14256) is not required and the resistivity of the water extract (see 3.2.6) shall be at least 45,000 ohm-centimeters.

4.2.4 Cleaning solvents. - Solvents to be used for the removal of grease, oil, and other dirt from the parts prior to soldering, as well as flux residues from the joint area, are divided into two groups as follows:

a) Non-flammable -

1) Tetrachloroethylene (perchloroethylene) - electronic grade.

2) Trichloroethylene - electronic grade.

3) Or any of the products approved by the procuring activity (see 6.3.2).

b) Flammable -

1) Ethyl alcohol conforming to O-E-760, Grade 1, class A or class B.

2) Isopropyl alcohol conforming to TT-I-735 - 99% pure.

4.2.4.1 Cleaning. - If parts and assembly can be totally submerged in the cleaning fluid without damage, this is the preferred method. In those cases where only brushing with the liquid cleaner is permissible, use only non-activated or mildly activated rosin core solder, where flux residues may be left in place. All residues from liquid fluxes should be removed.

4.2.5 Flexible insulation tubing or sleeving. - Except as otherwise required by the design, flexible electrical insulation tubing shall be either extruded vinyl plastic, polytetrafluoroethylene or heat shrinkable, as applicable. Polyvinyl chloride plastics shall not be used.

4.2.5.1 Extruded vinyl plastic tubing. - Extruded vinyl plastic tubing conforming to MIL-I-7444, Type I, may be used where exposure to temperatures in the range of -90° to +158°F will be encountered.

4.2.5.2 Tubing. - Polytetrafluoroethylene tubing which conforms to MIL-I-22129 may be used where exposure to high temperatures and high frequencies or both will be encountered. This tubing shall not be used in applications where the temperature exceeds 500°F. WARNING: When polytetrafluoroethylene resin compounds are heated above 400°F, small quantities of toxic vapors are emitted. Adequate exhaust ventilation shall be provided when polytetrafluoroethylene resin compounds are heated above this temperature.

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4.2.5.3 Heat shrinkable tubing. - Heat shrinkable tubing which conforms to MIL-I-23053 (Wep) may be used where shrink-fit tubing is required. Heat shrinkable tubing shall not be used where the application of heat will damage the soldered joint or component. Class 2, polytetrafluoroethylene heat shrinkable tubing shall not be used for this purpose.

4.2.6 Terminals. - Solder terminals shall readily accept solder according to MIL-STD 202, method 208 and require no presoldering preparation (e.g., tinning). Terminals may be plated and shall be provided with guide slots, holes, grooves, or similar features to ensure proper mating of the wire or wires with the terminals. Each terminal guide shall be located to facilitate application of solder. Terminals shall be of proper size to accommodate the conductors. Terminal slots and solder cups shall not be modified to accept oversize conductors, and conductors shall not be modified to accommodate undersize terminals. Terminals shall be stored in special sealed containers with a small bag of dessicant to prevent oxidation. The use of silver plated terminals on printed circuit boards shall not be permitted. Silicone base lubricants must not be applied on or near surfaces to be soldered.

4.2.7 Wire. - Wires shall conform to the applicable drawings and specifications and shall be capable of being stripped clean by means of the insulation strippers specified in 4.4.6. All wires, including lead wires, shall readily accept tinning and soldering, in accordance with MIL-STD-202, method 208.

4.2.8 Solderability. - The solderability of solid lead wires, terminals, and other termators which are normally joined by soft solder shall be determined in accordance with MIL-STD-202, method 208. They shall be tested in the "as received" condition and after aging according to paragraph 4.2, method 208.

4.3 Materials for soldering printed wiring. -

4.3.1 Materials. - Except when otherwise specified on detailed drawings, the materials utilized in soldering operations shall conform to the requirements of 4.3.2 through 4.3.6.

4.3.2 Solder. - The solder used shall be tin-lead conforming to composition Sn60 or Sn63 or Type S conforming to QQ-S-571.

4.3.3 Flux. - The soldering flux shall be noncorrosive and nonconductive rosin base flux (see 3) which has a melting point below the liquidus of solder. The flux shall have no harmful effect on the markings of components of the printed board and shall be readily removable with the solvent specified. The flux shall conform to MIL-F-14256 or be a product approved by the design activity. (See 6.3.3 for typical products which meet these requirements.)

4.3.4 Solvents. - An approved stabilized neutral, Ph 6.6 to 7.4, (as tested by a water extraction method) solvent shall be used which has the ability to degrease or remove rosin flux residue and other contaminates from printed circuit assemblies. Solvents shall conform to requirements of TT-I-735 Isopropyl Alcohol, Grade A, or be a product approved by the design activity. (See 6.3.2 for typical products which meet these requirements.)

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4.3.5 Materials and processes compatibility. - The Government provides no guarantee that materials and processes specified in this specification will not, in some combination, produce by-products or other undesirable incompatibilities which will render the finished product unacceptable. It shall be the responsibility of the manufacturer to select those processes and materials, which are compatible with one another and which best suit the end product desired.

4.3.6 Printed wiring board solderability. - The solderability of printed wiring boards shall be determined in accordance with MIL-P-55110, paragraph 4.9.11 or EIA Standard 319.

4.4 Equipment, tools, and application. -

4.4.1 Tools. - Except when otherwise specified, all tools utilized shall conform to the requirements of 4.4.2 through 4.4.11.

4.4.2 Brushes and steel wool. - Wire brushes and steel wool shall not be used. Medium stiff natural or synthetic bristle brushes may be used for cleaning.

4.4.3 Pliers. - Pliers used for cutting conductor wire and component leads shall be smooth (all serrations removed from plier jaw prior to use). They shall shear sharply and consistently produce a clean, smooth, cut surface along the entire cutting edge. Small, long-nose pliers or tweezers may be used for attaching or removing conductor wire and component leads.

4.4.4 Bending tools. - Bending and shaping of bare leads shall be accomplished with any tool including automatic bending tools which will not cut, nick, or in any way damage the leads or insulation. The lead shall be supported next to the component body. Needle nose pliers shall have rounded edges and shall not be serrated.

4.4.5 Clinching tools. - Clinching tools shall be made of a material which will not damage printed circuit conductors or component leads. The tools shall be used in such a manner as to prevent damage to printed circuit conductors or component leads.

4.4.6 Insulation strippers. -

4.4.6.1 Thermal strippers. - Thermal-type insulation strippers shall be used where practicable. (See Figure 1.) When required for personnel safety, an exhaust hood and fan ventilation system shall be used to exhaust toxic fumes when thermal stripping insulation, such as polytetrafluoroethylene or polyvinyl chloride. All excess or molten insulation must be removed after thermal stripping.

4.4.6.2 Precision cutting-type strippers. - Hand or die cutting-type strippers shall be used to strip glass braid and may be used in lieu of thermal strippers for removal of other insulation. Cutting-type strippers which permit operator adjustment shall not be used. Strippers shall be periodically checked for proper die hole size and operation and replaced when found defective. NOTE: Special die types are made for polytetrafluoroethylene, Kapton, or nylon jackets and should not be interchanged. (See Figure 1.)

4.4.7 Soldering iron tips. - Only iron clad tips shall be used.

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4.4.8 Soldering irons. - Soldering iron size (tip size and shape, voltage and wattage rating) and temperature shall be selected and controlled for optimum performance in relation to the work to be performed. (See 4.6.2.) Transformer-type solder guns shall not be used. Temperature controlled soldering irons are acceptable provided sensing of temperature is at the soldering iron tip.

4.4.9 Soldering iron holder. - A solder iron holder shall be provided for the soldering iron utilized. A cage type holder that leaves the soldering iron tip unsupported is preferred. The holder should be such that the iron handle is protected from rising heat. The holder should assist in maintaining proper iron temperature.

4.4.10 Soldering tools. - Tools shall not cut, nick, or in any way damage leads. Forked type tools generally referred to as soldering aids may be used, provided they are of the non-metallic type.

4.4.10.1 Thermal shunts. - Thermal shunts, or heat sinks, shall be used as necessary to protect heat-sensitive components such as semiconductors, crystal devices, meter movements, insulating materials, etc., from damage due to heat while soldering. Thermal shunts shall be of such material, size, shape, and design as to permit rapid application and removal with minimum interference to the soldering procedure and provide rapid heat removal from the area being soldered. Thermal shunts shall be held in place by suitable means, such as friction or spring tension, which will prevent damage to the surface and insulation of the wire and to the component being soldered. (See Figure 2.)

4.4.10.2 Anti-wicking tools. - Approved type anti-wicking tools (see 6.3), marked with conductor gage size, shall be used when required (See Figure 2.)

4.4.11 Tools and materials for cleaning. - Tools and materials for cleaning component leads, contact areas, gold plating, and soldering iron tips shall be as specified herein.

4.4.11.1 Component lead cleaning. - Component leads shall be cleaned with an efficient cleaning tool. Such a tool can be constructed utilizing 1/2-inch tinned copper shielding braid mounted in a spring-type tool as pictured in Figure 3. Knives, emery cloth, sandpaper, and other abrasives shall not be used.

4.4.11.2 Soldered areas. - A medium-stiff natural or synthetic bristle brush or a lint-free industrial cleansing tissue, dipped into an approved solvent, shall be used to remove excess flux following solder solidification. Wire brushes, knives, emery cloth, sandpaper, and other abrasives that produce a harsh abrasive action shall not be used.

4.4.11.3 Gold plating. - For removing gold plating from solder areas, use a pencil-style white typewriter eraser. It shall be furnished with a brush attachment.

4.4.11.4 Care of soldering iron tips. -

4.4.11.4.1 Unplated tips. - Shall not be used.

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4.4.11.4.2 Tinning plated tips. - Plated tips shall be tinned by scrubbing action on an alumina substrate (ceramic) while the iron is hot and immersed in a ball of solder and flux until a bright shiny solder coated surface is visible. This will not remove the plating. Files shall not be used for cleaning plated tips. Fine emery cloth may be used on the tip instead of the ceramic substrate.

4.4.11.4.3 Cleaning hot soldering iron tips. - A wet (utilize distilled water, not chlorinated drinking water), fine-texture sulfur-free natural or synthetic sponge shall be used by operators for cleaning heated soldering iron tips.

4.4.11.4.4 Tip maintenance. - The soldering iron tip shall always be checked for full insertion into the heating element and tight attachment to the iron. Oxidation scale shall not be allowed to accumulate between the heating element and the tip. The barrel of the cold iron shall be cleaned with a soft pipe cleaner or similar object when tip is removed, or once a week. A bright thin, but continuous, tinned surface shall be maintained on the tip working surface to ensure proper heat transfer and to avoid transfer of impurities to the solder connection.

4.4.11.5 Tool cleanliness. - Tool working surfaces shall be cleaned prior to use to remove all dirt, grease, flux, oil, and other foreign matter and shall be cleaned during use as necessary.

4.4.11.6 Handling. - Personnel handling cleaned wires, leads, terminals, and printed circuit boards shall wear thin, lintless, clean, white nylon gloves, or finger tabs to prevent contamination by contact with bare hands or fingers. (See 4.1.1.)

4.5 Preparation for soldering. -

4.5.1 Semiconductor soldering precautions. - Although transistors and diodes are in most cases sturdy for the job for which they are designed, there are some conditions that semiconductors will not tolerate. For example, the static electricity buildup in the body of a solderer or assembler while walking on a nonconductive surface may be sufficient to send a damaging pulse through a diode when discharged. In order to prevent such damage, persons coming into contact with or handling such devices should have a means of grounding themselves prior to touching or assembling the semiconductors. Only grounded soldering irons should be used for this application. Semiconductors are low-voltage devices, and as such cannot withstand high voltages; therefore, using personnel should periodically check their soldering irons for leakage voltage. The soldering iron should be replaced if there is enough leakage voltage present to damage a transistor or diode. It is also possible that such a condition may not immediately damage a semiconductor, but may contribute to a future breakdown. This damage may later manifest itself as a latent equipment failure. The manufacturer of the diodes or transistors can often provide leakage voltage tolerance limits. Some types of wire wrappers, automatic soldering machines, and similar inductively driven devices can develop transients which exceed the voltage which can be safely applied to semiconductors. When such devices are used in production or maintenance and if semiconductors come in contact with these tools, they must be lifted away before being turned on or off.

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4.5.2 Preparation of conductors. -

4.5.2.1 Correct lengths. - Wires and leads shall be cut to required lengths prior to attachment.

4.5.2.1.1 Solid hookup wire. - Solid hookup or bus wire shall not be used unless required by the design. When required, unsupported solid hookup or bus wire shall not exceed 5/8 inch in length between soldered connections. Wires which exceed this length shall be rigidly secured.

4.5.2.1.2 Stress relief. - All wires and leads terminated as a solder connection shall have sufficient slack in the form of a gradual bend. In applications where multiple wires are routed from a common cable trunk to equally spaced terminals, the stress relief shall be uniform to prevent stress on any one wire. (See Figure 4.)

4.5.2.1.3 Service loop. - Conductors routed from a harness or a cable to a terminal shall have sufficient slack in the form of a slight bend for one service loop. (See Figure 4.)

4.5.2.1.4 Component leads. - Leads shall have sufficient slack in the form of a slight bend to prevent strain at the terminal or component. Soldered components shall not impinge on adjacent circuitry. (See Figures 4 and 5.)

4.5.2.1.5 Mechanical support. - Solder joints shall not be subjected to mechanical loads. Mechanical support shall be provided, when component weight exceeds 1/4 ounce per load, for components or wire bundles by clamping, potting, embedding, or other means to prevent stresses on the solder joints.

4.5.2.1.6 Splicing. - The splicing of wires shall not be allowed unless specified on applicable drawings.

4.5.2.2 Insulation removal. - Insulation shall be removed from conductors by use of thermal strippers or precision cutting-type strippers, as applicable. (See 4.4.6.1 and 4.4.6.2.)

4.5.2.2.1 Mechanical stripper. - When using precision cutting-type strippers, the cutter shall be checked to ensure that the correct stripping hole is used for the corresponding wire size.

4.5.2.2.2 Arranging stranded wire lay. - Stranded wire shall be twisted in the direction of the lay during stripping operation in order to maintain its original form after stripping. The lay of the wire strands shall be restored if distributed, without bare finger contact.

4.5.2.2.3 Damage to insulation. - After stripping, the wire shall be examined for insulation damage. Wires with damaged insulation shall not be used.

4.5.2.2.4 Damage to wire. - After stripping, wire shall be examined with a magnifying glass not less than 6X nor greater than 10X to ensure that the wires have not been scratched, nicked, cut, scraped, broken, or otherwise damaged. Wires having any of these defects shall not be used. Stripping or trimming of wires shall not be allowed on the solder line. Every effort should be made to properly cut the wires to length, strip and tin prior to movement to assembly line.

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4.5.2.3 Insulation clearance. - The insulation shall not touch the solder joint surface. The bare wire shall extend from the solder joint not more than 1/8 inch or a distance equal to the outside diameter of the insulation of the wire, whichever is greater. (See 4.5.4 for measurement details and Figures 6 and 7.)

4.5.2.4 Tinning. - All portions of stranded wires which come in contact with the area to be soldered can be tinned by dipping the fluxed wire in a solder pot. (See Figure 8.) The tinning shall extend only far enough onto the wire to take full advantage of the area coming in contact with the connector or solder joint. A thin coating of solder shall be applied to all portions of the conductor wire coming in contact with the solder joint area. The solder shall penetrate to the inner strands of stranded conductors. To permit inspection for nicks or cuts at the point of insulation termination, solder and wicking shall not conceal the individual outer wire strands. A thermal shunt may be used to prevent this occurrence.

4.5.2.4.1 Wicking. - Wicking of solder up to the point of insulation termination is permitted provided the wicking does not obscure the contour of the wires. (See 4.6.4.8 and 5.3.7.2.) To permit inspection for nicks or cuts at the point of insulation termination, a thermal shunt shall be used to prevent solder wicking from obscuring the individual wire strands. (See 4.4.10.1 and 4.6.4.8.)

4.5.2.5 Cleaning and bending. - Prior to attachment to terminals and soldering, solid, tinned component leads shall be retinned, or cleaned with the tool specified in 4.4.11, until the tinned surface has a bright, shiny appearance. The component leads shall then be bent (with the tool specified in 4.4.4) into the form required for the connection to be made. When cleaning or bending welded leads, the lead shall be firmly held by a suitable tool on the side of the weld away from the component body during the cleaning or bending operation. The radius of the bend shall be equal to or greater than twice the lead diameter. The minimum distance from component and seal to the start of the bend shall be 1/16 inch. On components which have a welded lead, such as liquid electrolyte tantalum capacitors, the start of the bend shall be 1/16 inch or more from the weld. The bending action shall be accomplished by bending the lead end. The lead shall be supported next to the component body. (See Figure 9.)

4.5.3 Preparation of terminals. -

4.5.3.1 Cleaning of terminals. - Terminals shall be examined and where necessary, cleaned with an approved solvent (see 4.2.4) immediately prior to wire attachment and soldering. Oxidized terminals shall not be used.

4.5.3.2 Solder cups. - Solder cups shall be prefilled with sufficient solder to permit a proper solder joint. (See 4.6.3.2.)

4.5.4 Mechanical connection of conductors to terminals. -

4.5.4.1 Insulation tubing application. - Insulation tubing which conforms to the requirements of 4.2.5 shall be used for mechanical and electrical protection of soldered joints, wires, and leads and to protect wire harnesses and conductors against abrasion. Insulation tubing shall:

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a) Be installed over wires, leads, and harnesses prior to their attachment to terminals of relays, connectors, and similar items which are not protected by insulating grommets or potting;

b) Be pushed back far enough from the terminal so as not to interfere with the securing and soldering operations;

c) Be slipped back, after the solder has solidified and cooled and the joint cleaned, over the wires and terminals and, where applicable, heat shrunk.

After installation, the tubing shall extend above the stripped portion of the attached conductor a distance equal to or greater than the tubing diameter. (See Figure 10.)

4.5.4.2 Mechanical connections. - Conductors shall be connected to terminals as illustrated in Figures 11 through 17. The conductor shall be tangent to the terminal for the full curvature of the wrap or hook. The wrap of a conductor around a turret terminal shall not exceed 270° nor be less than 150°. Insulation shall not extend through or around any portion of the guide configuration on a terminal. (See 4.5.2.3.)

4.5.4.3 Multiple terminations. - The number of connections per terminal shall be in accordance with the drawing or specification. There shall be no overlapping of conductors. Each conductor shall be placed adjacent to the surface of the terminal. The wraps around a turret terminal may be clockwise or counterclockwise, but all wraps shall be in the same direction.

4.5.4.4 Turret terminals. - The bend to attach a wire or lead to a turret terminal shall be no smaller than 150° and no larger than 270°. The wires or leads shall be snugly attached to the base in the lower guide slot, or to the top shoulder in the upper guide slot. The end of the wire shall follow the radius and shall not protrude beyond the guide shoulders, depending on design considerations. The side route shall be used on all solid post turret-type terminals. Insulation clearance shall be referenced from the base or top shoulder, as applicable. (See Figures 11 and 12 and 4.5.2.3.)

4.5.4.5 Bifurcated terminals. -

4.5.4.5.1 Terminal fill. - Unless required by the design, terminal fill shall be as follows:

a) On bottom route and top route bifurcated terminals, a maximum of two conductors or wires shall be permitted for each route.

b) On side route terminals all conductors or wires shall be confined approximately within the lower 80% of the terminal. (See Figure 13.)

4.5.4.5.2 Bottom route. - Wires shall terminate with a 90° bend and shall be soldered to the terminal shoulder. The insulation clearance shall be measured from the point of entry of the wire into the terminal. (See Figure 14 and 4.5.2.3.)

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4.5.4.5.3 Side route. - The wire shall enter the mounting slot at a right angle and terminate with a 90° bend. When more than one wire is connected to the terminal, the direction of the 90° bend on each additional wire shall alternate. The first wire shall be soldered to the base and the vertical post. Additional wires shall be soldered as close as possible to the preceding wire, maintaining a clearance between the stranded wires equal to the thickness of the two insulations. The insulation on the first wire and all additional wires shall be a uniform distance from the terminal posts. Insulation clearance shall be referenced from the base. (See Figure 13 and 4.5.2.3.)

4.5.4.5.4 Top route. - A large diameter wire which fills the gap shall be inserted with no bend and shall require only fillets for retention. A tinned "filler" wire shall be used to hold in position a small diameter wire which does not fill the gap. As an alternate, a small diameter wire may be bent into a U-shape and inserted providing the combined diameter is sufficient to fill the gap. Insulation clearance shall be measured from the point of entry into the terminal. (See Figure 14 and 4.5.2.3.)

4.5.4.6 Hook, perforated, or pierced terminals. - The bend to attach wires and leads to hook, perforated, or pierced terminals shall be between 90° and 270°. Insulation clearance shall be as illustrated. (See Figures 15 and 16 and 4.5.2.3.)

4.5.4.7 Solder cup terminals. - The tinned wire or lead shall not be bent or formed for this application. All conductors terminating in a solder cup shall bottom in the cup. Insulation clearance shall be referenced from the point of entry into the cup. (See Figure 17 and 4.5.2.3.)

4.5.5 Shielded cable preparation. -

4.5.5.1 Protective sleeving. - Shielded cables, that are routed among exposed terminals and do not have an outer jacket of insulating material, shall be covered with sleeving. Unless otherwise specified on the applicable drawings or specifications, insulation sleeving conforming to MIL-I-7444 shall be used.

4.5.5.2 Shield removal for ground termination. - For shield ground terminations a portion of the shield shall be removed from the insulated conductor without damaging the individual strands of the shield. At a point approximately 1-1/2 inches back from the conductor end, mark the shield to indicate conductor breakout position. Loosen the shield by pushing it from the end toward the breakout point. Use a thin, bluntly-pointed tool to separate the loosened strands of the shield wire and make a hole large enough to pull the conductor wire through. The individual shield strands shall not be damaged by the tool. The shielding shall be pulled taut to smooth out any wrinkles of the shield. The loose end of the shield shall be twisted to insure a snug fit at the junction of the shield and insulation and to prevent the shield from creeping loose up the conductor. Shielding that has a ground termination shall not be grounded by any method that requires soldering on shielding that is positioned directly over or around the insulated conductor. (See Figure 18.)

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4.5.5.3 Termination of ungrounded shielding. - Braided shield that does not require a ground termination shall, at the point of termination, be insulated. The raw edge of the termination shall be insulated with insulation sleeving. The raw edge shall be folded or peeled back on the shield. The insulation sleeving shall be securely fitted over the termination, extending a minimum of 1/4 inch in both directions from the raw cut edge. (See Figure 19A.)

4.5.5.4 Termination of shielded single conductors in twisted pair or twisted shielded triple . - When terminating shielded single conductors of multi-conductor cable, the twisted shield shall be trimmed to approximately 1/2 inch from the breakout point and a piece of insulating tubing shall be applied over the shield and conductor insulation approximately 1/2 to 5/8 inch long covering the cut end of the shield and the shield breakout point. Sleeving MIL-I-23053 shall be shrunk to a tight fit. Breakout points may be staggered to prevent excessive diameter buildup of multiple conductor cables. (See Figure 19B.)

4.5.5.5 Termination of outer sheathed cables. - On outside type cables having a heavy outer sheath the overall shield shall be loosened by pushing back over the cable conductors. The shield shall then be folded back over the sheath approximately 1/2 to 5/8 inch. Cables will then have a piece of heat shrinkable insulating tubing applied over the folded back shielding and the sheath then shrunk in place. Tubing shall completely cover the shielding and extend approximately 1/4 inch beyond the cut end of the shield and 1/4 inch beyond the fold back point on the shield. (See Figure 20A.)

4.5.5.6 Termination of shields on cable harness. - Inside type cables or harness, having no outer sheath, will have a piece of heat shrinkable tubing applied over the wire bundle being approximately 3/4 inch back from the cut end of the shield and extending approximately 1/2 inch. The shield will then be loosened by pushing the shield back over the conductors. The loosened shield will then be folded back over the previously applied tubing and pulled taut. A second piece of insulating tubing will then be applied over the folded shield and extending from the fold point on the shield to at least 1/4 inch beyond both the previously applied tubing and the shielding. The tubing will be shrunk into place. (See Figure 20B.)

4.6 General soldering technique. (See 6.4.). -

4.6.1 Securing conductors. - The tinned wires and leads shall be formed and joined to the terminals as described in 4.5.4. Wires and leads shall be held snugly and rigidly to terminals in such a manner that there will be no motion relative to each other during the soldering operation and during cooling and solidification of the solder.

4.6.2 Heat application. - Apply the soldering iron tip to the connection area in such a manner that maximum heat will be transferred to the parts to be soldered and maximum protection will be afforded insulation and parts that will be adversely affected by excessive heat. Thermal shunts shall be used where required for the protection of insulation and heat sensitive parts. The connection area shall not be overheated. (See Figure 21A and the following table.)

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Prefered Soldering Iron Tip Temperatures

Flexible Printed Wire	475°F. min.
Printed Circuit Boards	475°F. min.
Terminals and Lugs	500°F. min.

4.6.3 Solder application. - Solder shall be applied to the joint when the temperature of the joint will readily melt the solder. Solder shall not be applied at the junction of the soldering iron tip and the parts to be joined, nor shall solder be melted on the soldering iron tip and allowed to flow over the parts to be joined. Solder shall cover the top of the conductor and a concave fillet shall be formed between the lower half of the conductor and the terminal. The contour of the terminal and the strands in the conductor shall not be completely obscured by solder. Forced cooling of the solder is prohibited. (See Figure 21B.)

4.6.3.1 All terminals except solder cup. - Sufficient solder shall be applied to form a slight concave fillet between the terminal and each side of the wire. The contour of the wire shall be visible after soldering. Excessive solder which completely obscures the wire and terminals is not permitted. (See Figures 22B and 24 through 27.)

4.6.3.2 Solder cup terminal. - The solder cup shall contain a sufficient amount of precut solder to completely fill the cup when the solder is melted and the tinned wire is inserted into the bottom of the cup. Sufficient heat shall be applied after the cup is filled with solder to assure that the flux has boiled up and out of the bottom of the cup. It is desired that continuous soldering iron contact be maintained throughout the soldering operation. The solder shall follow the contour of the cup entry slot and shall not spill over or adhere to the sides of the cup. Connection may be made with either a resistance or conduction type soldering iron. When a conduction type iron is used, the slight tinned effect occurring at the point where the tip contacts the base of the cup is normal and shall not be cause for rejection provided there are no peaks, globules, or excessive buildup of solder. Excess in the form of buildup or overflow on a hollow terminal must not be in excess as to cause a short circuit between pins of either conductors and excess solder cannot interfere with the mechanical function of the terminals. (See Figures 22C, 28, and 29.)

4.6.3.3 Solder coverage printed circuit. - The quantity of solder used shall be not more than is required to completely cover the joint area without completely obscuring the contour of the strands in the wire or the terminal. Solder fillets, where the parts being soldered join, shall taper gradually to a feather edge. (See Figures 22A, 30, and 31.)

4.6.3.4 Movement. - The parts being joined shall be held together in such a manner that the parts shall not move in relation to one another during the soldering operation nor after removing the soldering iron, until solder solidifies.

4.6.3.5 Cooling. - There shall be no forced cooling of the soldered joint. After soldering, the soldered joint shall not be disturbed until the solder has completely solidified and shall not be stressed until it has completely cooled.

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4.6.3.6 Reworking. - During soldering operations, a solder joint that is not initially satisfactory shall be allowed to cool to room temperature prior to the reapplication of heat, unless a heat sink is being utilized to dissipate the heat generated by the extended heat time. Connections which have been resoldered must meet the original soldering requirement.

4.6.3.7 Flux residues. - After the joint has cooled, all the flux shall be removed using a noncorrosive solvent as specified in 4.2.4.

4.6.3.8 Solder spatter. - There shall be no solder spatter on other surfaces.

4.6.4 Soldered connections appearance requirements. - See Figure 23.

4.6.4.1 Appearance. - The surface shall be neat and smooth, free of sharp points and foreign material.

4.6.4.2 Insufficient solder. - The solder shall completely cover the joint area.

4.6.4.3 Excessive solder. - The contour of the strands of stranded wire shall be visible. Beads of solder shall not project from the solder joint area. Solder shall not run down to the outside base of cup type terminals nor shall there be solder over any of the insulation. Whenever the contours of the wire or terminal are obscured by solder, the joint will be rejected as "excessive solder."

4.6.4.4 Rosin joint. - Flux shall not be trapped in the solder joint. The wire in the joint shall not move under pressure. The components or wires shall not be held together by rosin residues.

4.6.4.5 Cold joint. - The solder shall adhere smoothly to the parts joined. The joint shall not be chalky in appearance lacking metallic luster nor shall it have a rough piled-up surface.

4.6.4.6 Fractured joint. - There shall be no cracks in the solder or strain marks on the surface. The joint shall not present a rough, gritty appearance.

4.6.4.7 Wire termination. - The terminating end of a wire wrapped around a terminal shall not project more than 1/16 inch from the solder joint surface. Also, exposed copper shall not be visible under 6X to 10X magnification. Copper shall be allowed at wire ends when item is to be conformal coated (wire end shall not be considered part of the solder joint).

4.6.4.8 Capillary action (wicking). - The conductor leading to a soldered connection shall be flexible. The insulation shall not be distorted or split. Rigidity will not be cause for rejection when the area is to be potted or when the wire(s) leading to the area are secured to prevent excessive flexing.

4.6.4.9 Insulation. - The insulation shall not be charred, frayed, split, or pinched through exposing the conductor. The insulation shall not touch the solder joint surface.

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4.6.5 Removal of flux and impurities. - After the solder has solidified and cooled, all impurities shall be removed from each solder connection. (See 4.2.4.) The complete assembly shall be free of flux, residue, and impurities after cleaning. Handling of the finished product shall be kept to a minimum.

4.6.6 Connection inspection. - Prior to further processing, the quality of each soldered connection shall be determined by visual inspection, performed in conjunction with appropriate commodity drawings and this specification. Suspected defects will be verified by magnification using a magnifying glass not less than 6X nor greater than 10X. Reliable, soldered, electrical connections shall be clean and have a shiny, bright appearance; no porosity; good fillet between conductor and terminal, good adherence of conductor to terminal; and no excess solder. In all applications except for solder cups, the contour of the wire or lead shall be visible after soldering and the end of the wire or lead shall not extend beyond terminal dimensions. In solder cup applications, the contour of the wires and leads shall be visible from the insulation to the point of entry into the cup. Wires shall not be bent more than 45° when inspecting for capillary action, etc., nor shall wires be bent unnecessarily; since severe and repetitious bending could result in a broken wire under the insulation.

4.6.6.1 Mark of acceptance. - Soldered connections that comply with the requirements may be identified by color coding as specified in 5.2.1.1.

4.6.7 Reworking unsatisfactory connections. -

4.6.7.1 General. - Solder connections which do not meet the criteria specified in this standard shall be reworked by removing the solder, cleaning the connection, and resoldering.

4.6.7.2 Solder removal by wicking. - Solder shall be removed only by wicking (see Figure 32) using stranded wire (19 strand is preferable) or shielding braid, as follows:

a) Strip approximately 1/2 inch of insulation from the wire.

b) Dip the bare wire in liquid flux, according to MIL-F-14256.

c) Place the fluxed wire on the solder connection and place the hot soldering iron tip on the wire, and;

d) Remove the tip and wire simultaneously as soon as the desired amount of solder has wicked onto the wire.

e) Clean surfaces per 4.4.11.2 to restore solderability.

4.6.7.3 Resoldering. - After the solder has been removed, the wires and leads shall be resoldered in accordance with the applicable portions of 4.6.3.

4.6.7.4 Connection reinspection. - Each resoldered connection shall be inspected in accordance with 4.6.6.

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4.7 Soldering of printed circuits. (See 6.4.). -

4.7.1 Solderability. - Conductor patterns of printed wiring shall be readily solderable. Prior to issuing printed wiring boards to soldering operators, solderability shall be ascertained in accordance with applicable specifications. (See 4.3.6.)

4.7.2 Protection during assembly. - During assembly and inspection, the printed wiring shall be suitably protected to prevent deformation or damage.

4.7.3 Preassembly quality requirements. - A visual examination of each printed wiring shall be made prior to assembly. Operators shall not assemble those boards having any of the defects defined in 5.4.2. (See Figures 33 and 34.)

4.7.4 Eyelets and tubelets. - Eyelets and tubelets shall not be used on printed wiring for through connections unless specified on the applicable drawing.

4.7.5 Terminal mounting. - Printed circuit pads from which component leads may frequently be removed shall be reinforced with terminals.

4.7.5.1 Hole forming. - The pad shall be drilled, punched, or reamed, as applicable, to a diameter that will permit the terminal shank to be pressed through the board by hand. A press fit is not necessary but the terminal shall not fall out when the board is inverted.

4.7.5.2 Insertion and alinement. - The terminal shank shall be pressed through the board and the terminal alined as shown in Figure 35A. A 360° contact shall be maintained between the mounting shoulder of the terminal and the board.

4.7.5.3 Terminal swaging. - Printed circuit pads which will accept terminals shall not be gold plated and shall be tinned.

4.7.5.3.1 General. - A V-type (funnel) swage shall be used on the printed wiring where the swaged end of the terminal terminates in a circuit pad or pattern. A roll-type swage shall be used only where the swaged end terminates directly to the printed circuit board base material. After swaging, the roll or flare shall not be split and shall not abrade or damage the circuit pattern. Solder rings with rosin flux per MIL-F-14256 are recommended for use on swaged terminals to provide a uniform and reliable solder connection.

4.7.5.3.2 Swaging. - When solder rings are used, the ring shall be placed over the terminal shank before swaging. A V-type (funnel) swage shall be made on the terminal, using a swaging tool as shown in Figure 35B. The point of the V-swage shall enter the terminal shank only far enough to produce a handtight fit of the terminal, see Figure 36A.

4.7.5.4 Terminal soldering. -

4.7.5.4.1 Using modified soldering iron tip. - When using a modified soldering iron tip, as shown in Figure 36B, the soldering tip shall be placed over the terminal head and rosin core solder applied to the area where the terminal shank and pad intersect (if a solder ring is not used). The solder shall be allowed to flow until a proper connection is obtained. The soldering iron tip shall then be removed from the terminal head.

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4.7.5.4.2 Using standard soldering iron tip. - When using a standard conical soldering iron tip, the tip shall be placed in the barrel of the terminal as shown in Figure 36B, and soldered as stated in 4.7.5.4.1.

4.7.6 Mounting and soldering of component parts. -

4.7.6.1 Insulation tubing for parts. -

4.7.6.1.1 Glass encased components. - When using epoxy potting compound to encapsulate glass encased components (e.g., diodes) the components shall be enclosed in tubing conforming to 4.2.5, prior to mounting and encapsulation for protection against thermal expansion.

4.7.6.1.2 Metal case components. - All metal case components mounted over printed conductor wiring shall be insulated with transparent tubing conforming to the criteria set forth in 4.2.5. (See Figure 42.) These components shall not be mounted over solder connections.

4.7.6.2 Lead bending. - Component leads shall be cleaned and bent per 4.5.2.5.

4.7.6.3 Component mounting. - Components shall be mounted parallel to and on the side opposite the printed circuitry and in contact with the printed circuit board, unless otherwise specified by the design. All component parts shall be potted, embedded, or supported by a suitable method. Component leads and other conductors terminated directly at the circuit pad shall extend through the board approximately the length of the pad radius and approximately a maximum of the pad diameter and shall be clinched in the direction of, whenever possible, and in contact with the circuit pattern. (See Figures 37, 38, 39, and 40A.)

When the lead is longer than a pad radius, it shall always be clinched within the printed circuit path. (See Figure 46.) The component leads and other wires shall not extend out beyond the circuit pattern.

4.7.6.4 Special mounting. - Component leads which cannot be bent and clinched flush with the circuit shall be properly cleaned and cut to a length that will permit the lead to extend 1/32 inch above the solder pad. (See Figure 40B.) Unless otherwise required by the design, components shall be rigidly secured to the printed circuit board by a suitable mounting clamp. Lead solderability of components mounted in this manner must be proven by documented test results using written test procedures. To minimize special mounting, components designed for printed circuit application shall be used when obtainable.

4.7.6.5 Boards with circuit pattern on both sides. - Component leads that interconnect two-sided circuitry shall show solder wetting and filleting on both sides. (See Figure 41.) To ensure the reliability of the solder connection, solder first from the bottom side trying to solder through the whole fillet including the top. If the fillet is not perfect on the topside, wait until the joint cools and resolder from the top to obtain the proper fillet. (See tip temperature, 4.6.2.) This procedure also applys for multilayer boards.

4.7.7 Soldering to gold plated circuits. - Soldering to printed circuit pads shall be accomplished only on the base metal or on tin-lead plated/fused tin lead alloy base metal. When the printed circuit pad is gold plated, the gold plating shall be removed from the pad using the eraser specified in 4.4.11.3 or the wicking method described in 4.6.7.2.

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Sufficient pressure shall be used to remove only the gold plating from the pad. Eraser particles shall be removed by brushing lightly with the eraser brush.

4.7.8 Post soldering insepction of terminals and boards. - Each completed printed circuit board shall be inspected using the criteria outlined in 4.3.6 and 4.6.6 for typical satisfactory and unsatisfactory connections. (See Figures 43 to 46.)

4.7.9 Conformal coating. - Unless otherwise required by the design, printed wiring shall receive a clean, transparent, insulative protective coating on both sides in accordance with applicable drawings for polyurethane coating after wiring, functional checkout, and acceptance by supplier inspection.

4.7.10 Protective packaging. - The printed wiring shall be protected between assembly operations by storing it in a protective container made of a material which does not emit gases or chemicals that could be detrimental to the printed circuitry.

4.8 Training and certification. - The contractor is responsible for:

4.8.1 General requirements. - Providing necessary training of his personnel in the use of equipment employed and for insuring that all personnel who perform or inspect soldering are adequately skilled to fabricate the types of soldered connections required.

4.8.2 Records. - Maintaining appropriate records of training, including the certification criteria for each individual.

4.8.3 Certification. - Certifying all personnel who perform or inspect soldering as being currently qualified to fulfill all requirements of this specification pertaining to the types of connections to be soldered. Evidence of certification status shall be posted in the work area and each person shall carry a card indicating his current certification status.

4.8.4 Maintenance of certification status. -

4.8.4.1 The procuring installation or its designated representative, or the contractor's instructor, may require contractor soldering personnel to demonstrate proficient workmanship on applicable hardware, or to be recertified.

4.8.4.2 The procuring installation or its designated representative or the contractor's instructor, may require contractor inspection personnel to demonstrate proficient inspection performance and knowledge on applicable hardware or to be recertified.

4.8.5 Recertification. -

4.8.5.1 The need for recertification shall be based on observation of the unsatisfactory quality of articles fabricated, or interruption of the work period for more than 90 calendar days.

4.8.5.2 Recertification shall be required when:

a) New techniques have been developed which require different skills.

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b) Certificate holder changes employment.

c) There is reason to question the inspector's performance.

4.8.5.3 Procedures for recertification shall include sufficient training or retraining to enable the candidate to demonstrate proficiency in fabricating or inspecting the types of solder connections involved in his assigned work. A proficiency demonstration shall be required of each candidate.

4.8.6 Revocation of certified status. - Certifications shall be revoked for operators or inspectors when:

a) Certificate holder requires recertification according to 4.8.5.2 and fails to be recertified.

b) Certificate holder fails to meet visual acuity requirements of 4.8.7.

4.8.7 Vision requirements. -

4.8.7.1 The contractor is responsible for ensuring that all personnel who perform soldering or inspect soldered connections meet the following vision test requirements as a prerequisite to training, certification, and recertification. The vision requirements may be met with corrected vision, i.e., eye-glasses. The eye tests shall be administered by qualified personnel, using standard instruments and techniques. Results of the vision examination shall be maintained and available for review.

4.8.7.2 The following are the minimum vision requirements:

a) Far vision. - Snellen Chart 20/50.

b) Near vision. - Jaeger 1 at 14 inches; or reduced Snellen 20/20 or equivalent.

c) Color vision. - Ability to distinguish red, green, blue, and yellow colors as prescribed in Dvorine Charts, Ishihara Plates, or AOD-HER Tests. A practical test, using color coded wires and/or color coded electrical parts as applicable, will be acceptable for color vision testing.

4.8.8 Visual standards. - The contractor shall:

4.8.8.1 Prepare visual workmanship standards consisting of satisfactory work samples or visual aids which clearly illustrate the quality characteristics of all soldered connections involved.

4.8.8.2 Utilize applicable illustrations in this standard supplemented as necessary for visual standards.

4.8.8.3 For approved connections, other than those illustrated in this standard, prepare appropriate visual standards.

4.8.8.4 Clearly illustrate by these standards preferred workmanship and the difference between acceptable and unacceptable workmanship.

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4.8.8.5 Make applicable visual standards readily available to concerned personnel and use them in the training program.

4.8.8.6 Use these standards for inspection criteria and evaluation of personnel performance.

4.9 Workmanship. - Equipment which has been soldered shall comply with the performance requirements of applicable detail specifications insofar as soldering operations are involved. The circuitry shall be free of dirt, chips, and other foreign material, and, whenever possible, free of rosin flux residues. Conductors shall be dressed together neatly and parallel to each other. Components and insulation shall exhibit no damage as a result of the soldering operation. Refer to the Appendix for visual standards.

5. DETAIL REQUIREMENTS

5.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the standard where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

5.1.1 General. - The contractor is responsible for maintaining a working area (see 4.1) and a documented soldering program which meets the requirements of this standard for the types of solder connections specified by the drawings referenced in the contract. Portions of this standard, including illustrations, may be abstracted for the program.

5.1.1.1 Document submittal. -

5.1.1.1.1 Documents required herein shall be submitted to the procuring activity or its designated representative for review and approval. Applicable contractor soldering program documents, or portions thereof, accepted on other government contracts shall be included to avoid duplication of effort.

5.1.1.1.2 The contractor shall establish, document, and maintain an effective training program to certify and recertify all personnel who will perform hand soldering operations to satisfy the requirements for the type of soldering connections required by the contract design. The program shall include:

a) Qualification of Instructors - To qualify as an instructor the individual shall have completed an 80-hour instructor's course at an appro ved school.

b) Procedures for training.

c) Lesson plan(s).

d) Hours of instruction.

e) Procedures for certification and recertification.

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5.1.1.1.3 The contractor shall develop, document, and maintain a feedback system that will provide data with regards to defective solder conditions for use in the recertification program. The system developed shall be submitted to the procuring activity for approval prior to use.

5.2 Inspection. -

5.2.1 Visual. - One hundred percent visual inspection of all soldered connections shall be performed in conjunction with appropriate commodity drawings and this standard. A magnifying glass of no greater magnification than 10X nor less than 6X shall be used to aid visual inspection. Deviation from any of the requirements shall be cause for rejection.

5.2.1.1 Mark of acceptance. - Soldered connections that comply with the requirements (where possible) may be identified by color coding with a nonfungus nutrient ink or paint that shall not affect the electrical characteristics of the assembled circuitry. The coding shall be of contrasting color to the area it is applied. When applied, it shall be limited in size but readily visible. For post type terminal connections, the coding shall be applied on the head of the terminal. Where receptacle or cup type terminals are involved, the coding shall not be applied on the solder joint.

5.2.2 Mechanical. - A mechanical check of a soldered joint is acceptable only when required to supplement visual inspection. After a mechanical check is performed, the joint must be reworked, if required, to meet the requirements of this standard. Connections which are reworked must meet the original soldering requirement.

5.2.3 Surveillance . - A surveillance program will be conducted to determine that compliance with Section 4 of this standard is being maintained, and to observe the control and disposition of nonconforming material. This includes periodic inspection of the work area, tools, materials, and procedures.

5.3 Acceptance criteria. - All soldered connections shall be inspected for conformance to the following conditions.

5.3.1 Solder connection. - The solder connection shall have a smooth, bright appearance, without porosity, cracks, pits that bottom out, or surface strain lines. Solder shall cover the top of the conductor wire and concave fillet should be observed between the lower half of the conductor wire and the terminal. There shall be no foreign material or threads of insulation embedded in the solder. (See Figure 22.)

5.3.2 Residual flux. - Each soldered connection shall be inspected to determine compliance with 4.6.5.

5.3.3 Excessive solder. - Beads of solder shall not project from the terminal and no solder shall exist as runs on the outside of the terminal. There shall be no solder splatter on the adjacent components or surfaces. (See Figure 23.) Solder is excessive if fillet does not reveal the outline of soldered components as specified in 4.6.3.

5.3.4 Cold solder joint. The solder shall adhere firmly and smoothly to the parts joined. The joint shall not be chalky in appearance, lacking metallic luster, nor shall it have a rough, gritty, piled-up surface. (See Figure 23.)

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5.3.5 Rosin joint. - Flux shall not be trapped in the solder joint, nor shall rosin flux residues hold the components together.

5.3.6 Insulation. - The insulation shall not be charred, frayed, split, or pinched through exposing the conductor wire. Slight discoloration of insulation shall not be considered cause for rejection (See 4.5.2.2.3.) Insulation clearance shall be in accordance with 4.5.2.3. (See Figure 6.)

5.3.7 Conductor wire. - The wire shall not be ringed, nicked, or cut; nor shall there be any unsoldered strands. Wire diameter shall not be reduced. Exposed copper ends on the wire shall be completely covered with solder unless conformal coated. (See 4.5.2.2.4.)

5.3.7.1 Conductor connection. - Each soldered connection shall be inspected to determine compliance with 4.6.1 and 4.6.6.

5.3.7.2 Capillary action (wicking). - Wick length of solder on the wire strands shall be visible and shall not extend into the insulation as specified in 4.5.2.4.1 and 4.6.4.8.

5.3.8 Conductor wire and component lead tension. - All conductor wires and component leads going to a soldered connection shall have slack in the form of an arc or gradual bend. (See Figure 4 and 4.5.2.1.2.) Soldered components shall not be in electrical contact with adjacent circuitry, or other components.

5.3.9 Excessive heat. - Components and insulation shall not show evidence of excessive heat. (See 4.6.2.)

5.3.10 Shielding. - All shielding shall be in accordance with 4.5.5.1.

5.3.11 Multiple terminations. - The number of connections per terminal shall be in accordance with the drawing or specification. Each conductor wire shall be adjacent to the surface of the terminal, not one overlapping another. Multiple connections on turrent type terminals shall be in accordance with 4.5.4.3. (See Figure 11.)

5.4 Rejection Criteria. -

5.4.1 General Criteria. - Evidence of any defect, including but not limited to the following, shall be cause for rejection:

a) Charring, burning, or other damage to insulation

b) Splattering of flux or solder on adjacent connections or components

c) Solder points (peaks)

d) Pits, scars, or holes

e) Excessive solder which obscures the connection configuration

f) Excessive wicking. (See 4.5.2.4.1, 4.6.4.8, and 5.3.7.2.)

g) Loose leads or wires

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h) Cold solder connections

i) Rosin solder connection

j) Fractured solder connection

k) Cut, nicked, or scraped leads or wires; insufficient slack

l) Unclean connection (e.g., lint, residue, flux, solder, splash, dirt, etc.)

m) Dewetting

n) Insufficient solder

o) Visible bare primary conductor within the solder joint area

p) Clinched leads resulting in a reduction of the required spacing between conductors

q) Birdcaging. (See Figure 12.)

r) Splicing. (See 4.5.2.1.6.)

s) Plated through holes not filled with continuous solder plug

t) Pads connected by plated through holes and eyelets connecting pads on multi-layer boards or double sided printed circuit boards show evidence of failure to wet the metallic surfaces. See comments h and s above

u) Use of unapproved tools or improper use of approved tools

v) Work area environmental controls improper

w) Use of improper materials

x) Solderers not certified

5.4.2 Printed wiring criteria. - In addition to those criteria specified in 5.4.1, evidence of any defects including but not limited to the following, shall be cause for rejection of printed wiring:

a) Pits, scratches, pinholes, or undercutting that reduce the conductor cross-sectional area more than 20%

b) Separation of the conductor pattern from the base laminate

c) Blisters in the conductor pattern

d) Delamination of the base material

e) Wrinkles in the conductor pattern

f) Dirt, grease, or other foreign matter on the printed wiring

g) Scratched, abraded, or scraped finish that will change the electrical resistance

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5.5 Rejects. - Defects shall be tagged and the type of deviation noted on the tag.

5.6 Rework. - Soldered joints, reworked to correct deficiencies, shall be one hundred percent reinspected for the deficiencies in question and for any other features that may have been affected by the reworking operation.

5.7 Inspection standards. - Figures 1 through 46 and any additional inspection provisions provided or approved by the procuring activity shall be used to aid in determining compliance with the requirements set fourth in this standard.

5.8 Qualification and certification. - The qualification and certification of soldering operators and soldering inspectors when required, (see 5.1.1.2), shall be in accordance with 4.8.

5.9 Work areas and environments. - Work areas and environments approval, when required (see 6.2), shall conform to 4.1.

6. NOTES

6.1 Intended use. - It is intended that the soldering procedures and techniques covered by this standard are to be utilized for the soldering of highly reliable electrical connections used for electrical and electronic equipment of military hardware and systems.

6.2 Ordering data. -

a) Title, number, and date of this military standard

b) If certification (see 4.8 and 5.8) of solderers and inspectors per this standard is not required, the appropriate contractual document shall indicate same.

c) If the work area requirements (see 4.1) of this standard are not required, the appropriate contractual document shall indicate same.

6.3 Approved products. - This list of approved products has been prepared for use by or for the Government during actual soldering operations covered by this standard. This listing is not intended to and does not signify endorsement of the product by the Department of Defense. All products listed herein have been qualified under actual use during actual soldering operations as specified in the latest effective issue of this standard. This listing is subject to change without notice; revisions or amendments to this listing will be issued as necessary.

Thermal Strippers, Manual (see 4.4.6.1): Contact, Inc. (hot tip) No. II, 101-CD-W515 or equivalent.

Mechanical Strippers (see 4.4.6.2): Ideal 45-173, 45-174, 45-183, and 45-184 or equivalent.

Soldering Irons (see 4.4.8): Hexagon P 2535, Weller WTCP-L, Weller WMPC or equivalent.

Resistance Soldering Apparatus (see 4.4.8): Contact, Inc. Model H-101-CD-T-103 or equivalent.

Anti-Wicking tools (see 4.4.10.2): Clauss or equivalent.

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6.3.1 Solvents. -

EI Dupont de Nemours & Co., Inc. freon T-E 35 solvent.

Alpha Metals Co., No. 563 cleaner or 1001 cleaner

6.3.2 Flux.

Kester Solder No. 197 with No. 108 thinner
Alpha Metals No. 611 with No. 412 thinner

6.4 Soldering methods. - The use of soldering methods other than hand soldering is acceptable under this military standard provided the same quality standards are applied to the finished product and the finished product complies with the requirements specified herein. [See MIL-S-46844A(MI).]

6.5 Supersession data. - This military standard supersedes and includes the requirements of MIL-S-60902(MU), dated 17 August 1967.

CUSTODIAN:
Army (MU)

PREPARING ACTIVITY:
Army (MU)

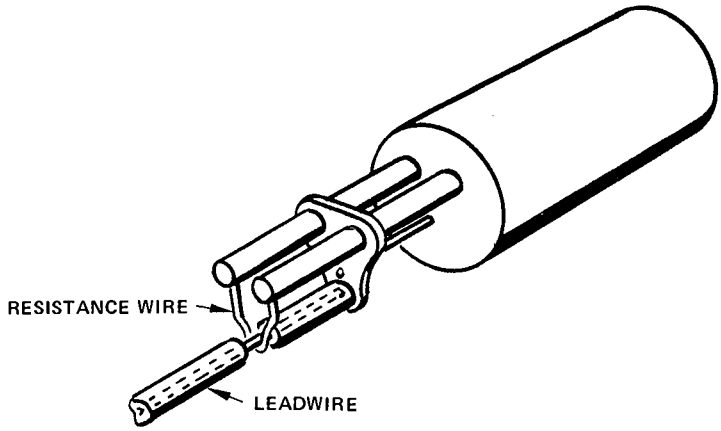
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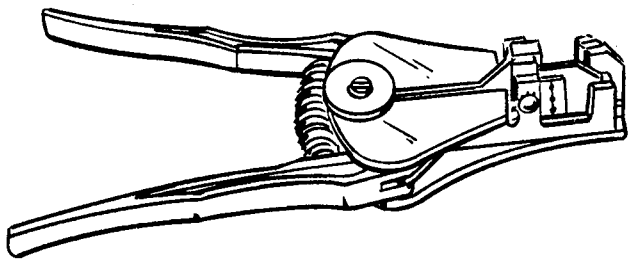
APPENDIX

ILLUSTRATIONS AND VISUAL STANDARDS

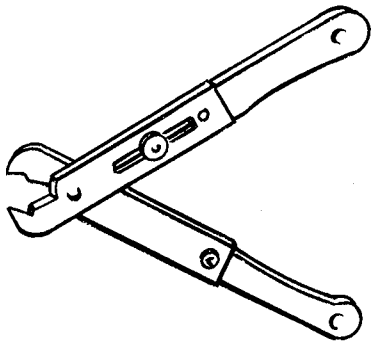
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TYPICAL THERMAL STRIPPER (PARA 4.4.6.1)



ONE STYLE OF MECHANICAL HAND WIRE STRIPPER (PARA 4.4.6.2)



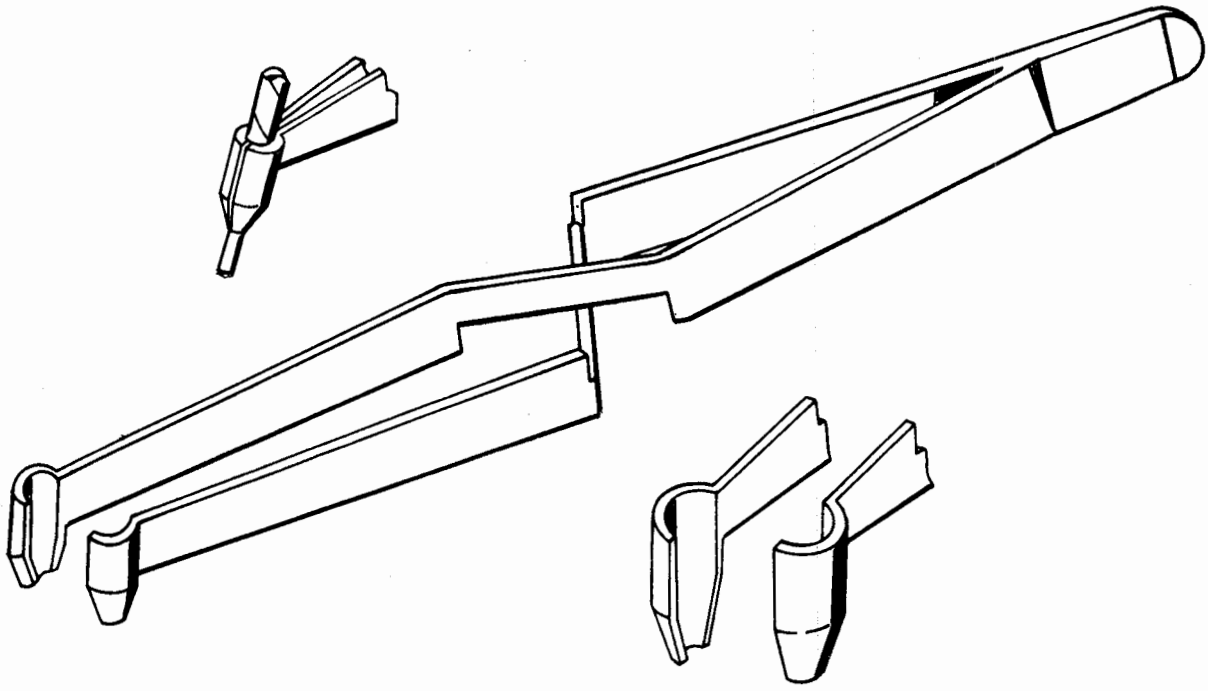
TYPICAL IMPROPER CUTTING-TYPE STRIPPER, NOT PERMISSIBLE TO USE (PARA 4.4.6.2)

FIGURE 1. STRIPPING TOOLS

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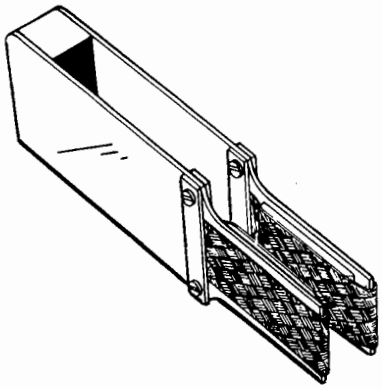
TYPICAL THERMAL SHUNTS (PARA 4.4.10.1)



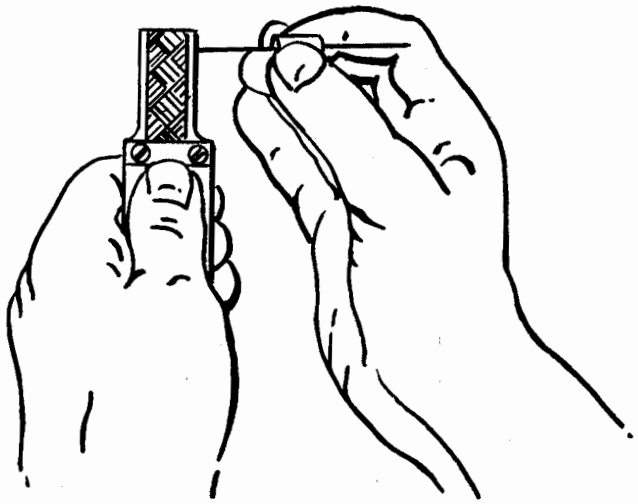
ANTI-WICKING TOOLS (PARA 4.4.10.2)

FIGURE 2. THERMAL SHUNTS AND ANTI-WICKING TOOLS

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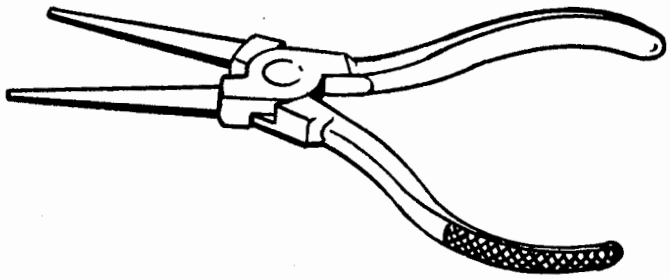


TYPICAL COMPONENT
LEAD CLEANING TOOL



LEAD CLEANING OPERATION

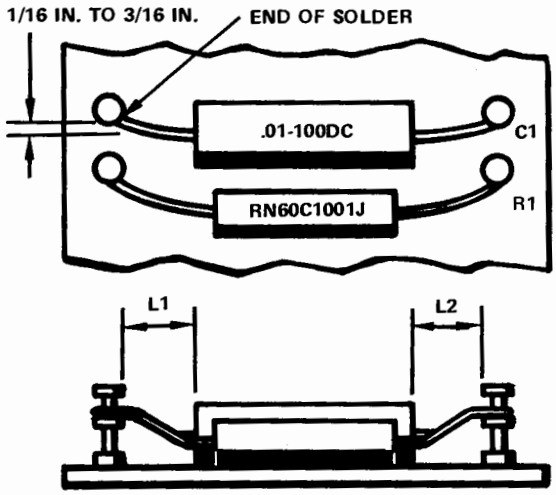
LEAD CLEANING TOOL (PARA 4.4.11.1)



NEEDLE NOSE PLIERS WITH ROUNDED JAWS FOR FORMING LEADS (PARA 4.4.4)

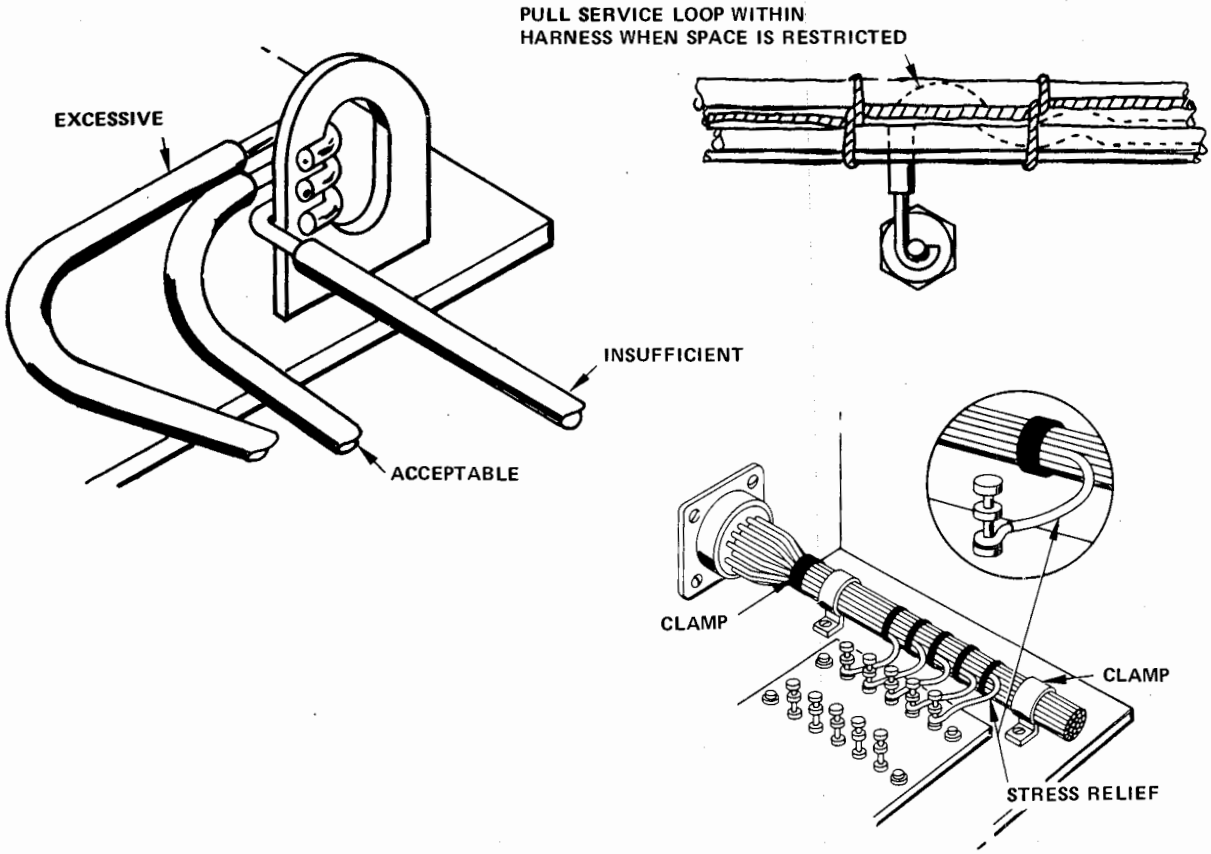
FIGURE 3. LEAD CLEANING AND BENDING TOOLS

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1. DIFFERENCE BETWEEN L1 AND L2 IS 1/8 INCH, OR LESS.
2. PART OFFSET TO BE IN LINE WITH REFERENCE DESIGNATION (APPROX. 1/8 IN.).
3. PART SUPPORTED BY MOUNTING SURFACE (UNLESS OTHERWISE SPECIFIED ON DRAWING) WITHIN 1/32 IN.
4. VALUE VISIBLE ON TOP OF PART, IF PRACTICAL.
5. MINIMUM DISTANCE TO START OF BEND IS 2 LEAD DIAMETERS.

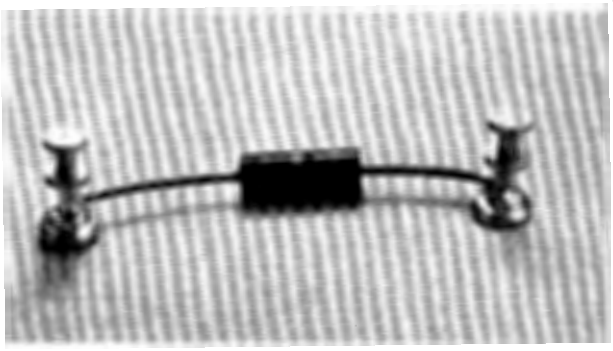
STRESS RELIEF IN COMPONENT MOUNTING (PARA 4.5.2.1.4)



STRESS RELIEF IN WIRING (PARA 4.5.2.1.3)

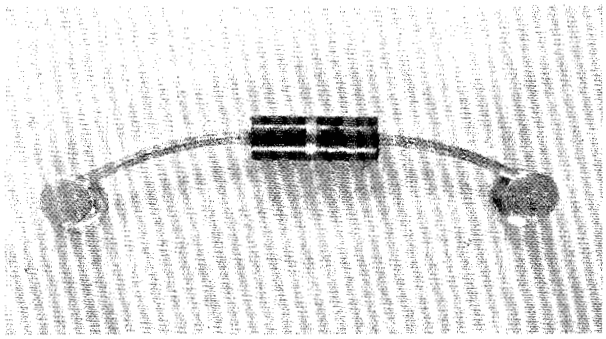
FIGURE 4. STRESS RELIEF

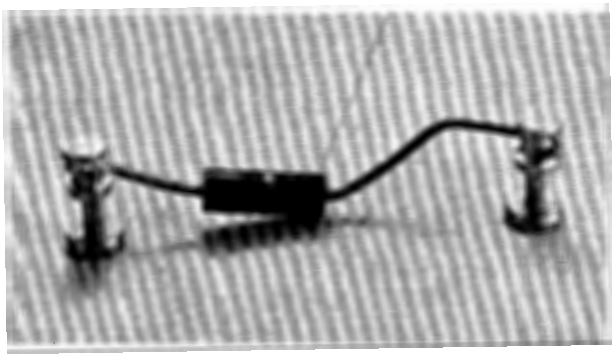
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DESIRED

1. COMPONENT CENTERED BETWEEN TERMINALS AND SEATED FLUSH ON BOARDS.
2. LEADS NOT UNDER TENSION – COMPONENT LEADS FORM A SLIGHT LOOP TO OFFSET COMPONENT BODY FROM TERMINAL CENTERLINE.
3. LEAD BEND HAS BEEN MADE AT LEAST 1/8 INCH FROM COMPONENT BODY TO PROTECT COMPONENT AND LEAD WELD.





REJECT

1. COMPONENT NOT CENTERED BETWEEN TERMINALS.
2. COMPONENT NOT FLUSH WITH BOARD.
3. LEAD BENT AT COMPONENT BODY.
4. LEADS NOT SOLDERED AT BASE OF TERMINAL.
5. VERY POOR QUALITY THROUGHOUT INSTALLATION.

FIGURE 5. COMPONENT STRESS RELIEF (PARA 4.5.2.1.4)

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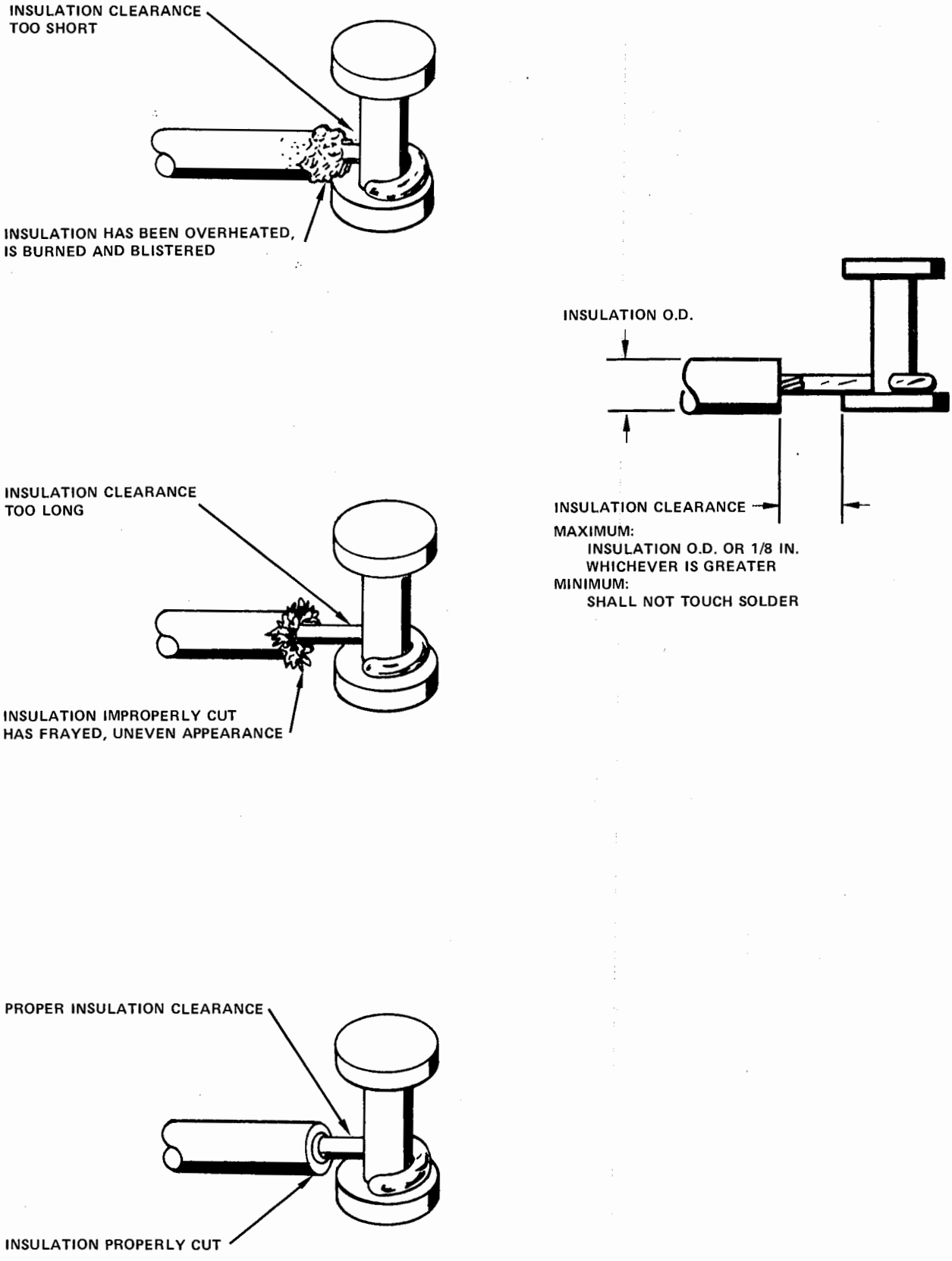


FIGURE 6. INSULATION TRIM (PARA 4.5.2.3)

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UNACCEPTABLE



INSULATION CUT BACK TOO FAR WITH SMALL DIAMETER WIRE

DESIRED



CORRECT TRIM WITH SMALL DIAMETER WIRE



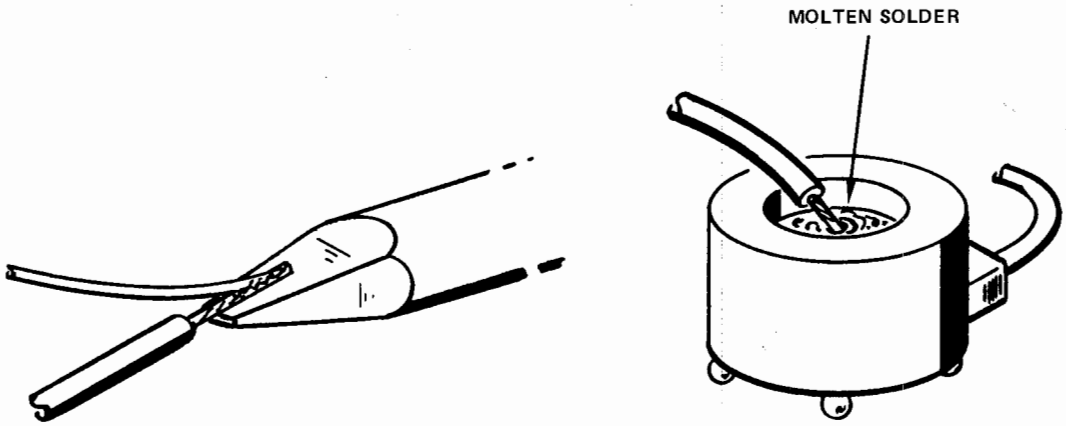
INSULATION NOT CUT BACK FAR ENOUGH WITH LARGE DIAMETER WIRE



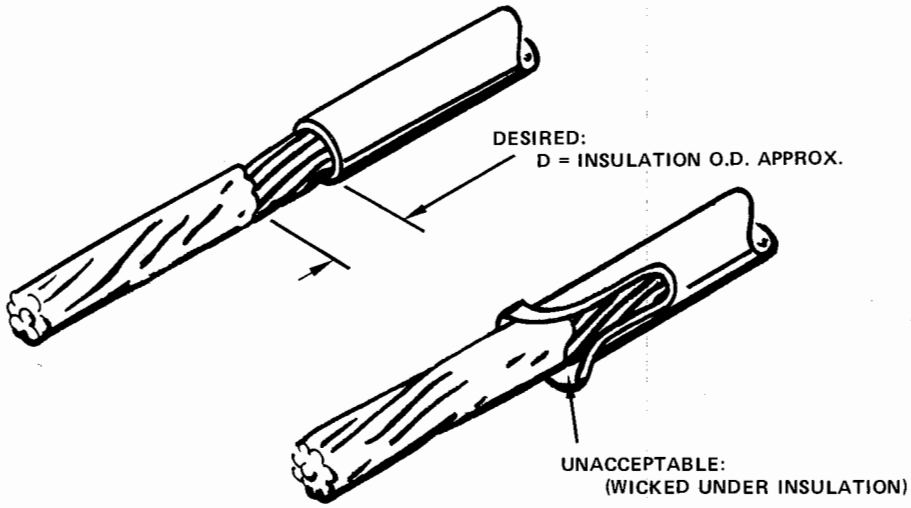
CORRECT TRIM WITH LARGE DIAMETER WIRE

FIGURE 7. INSULATION TRIM (PARA 4.5.2.3)

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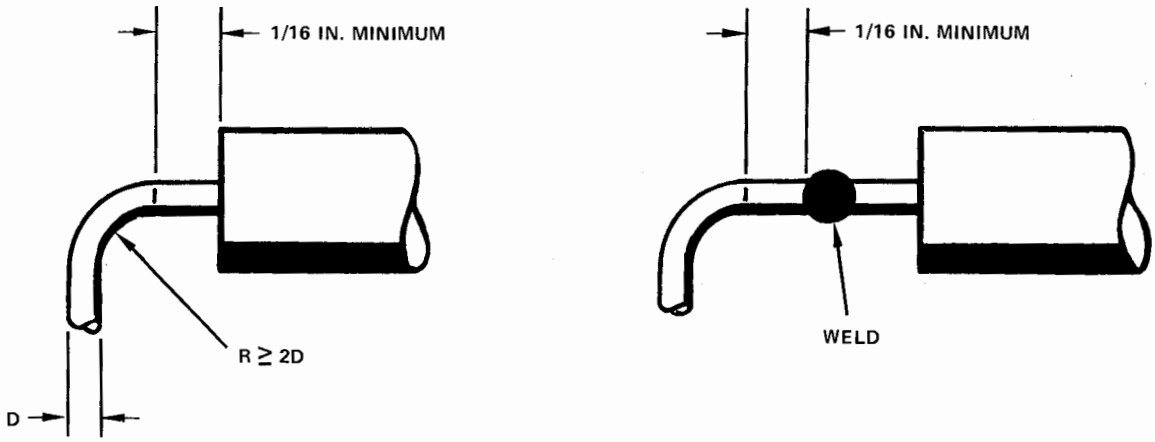
SUGGESTED METHODS FOR TINNING (WIRE)



EXAMPLES OF TINNED WIRE

FIGURE 8. WIRE TINNING (PARA 4.5.2.4)

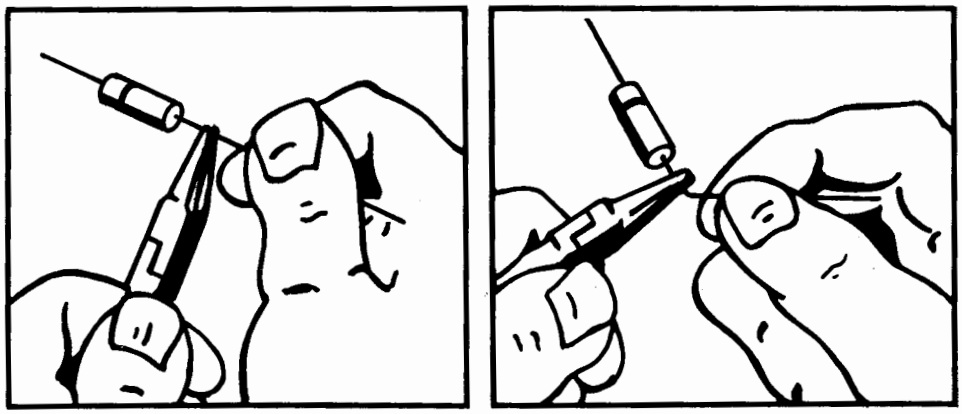
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STANDARD LEAD

WELDED LEAD

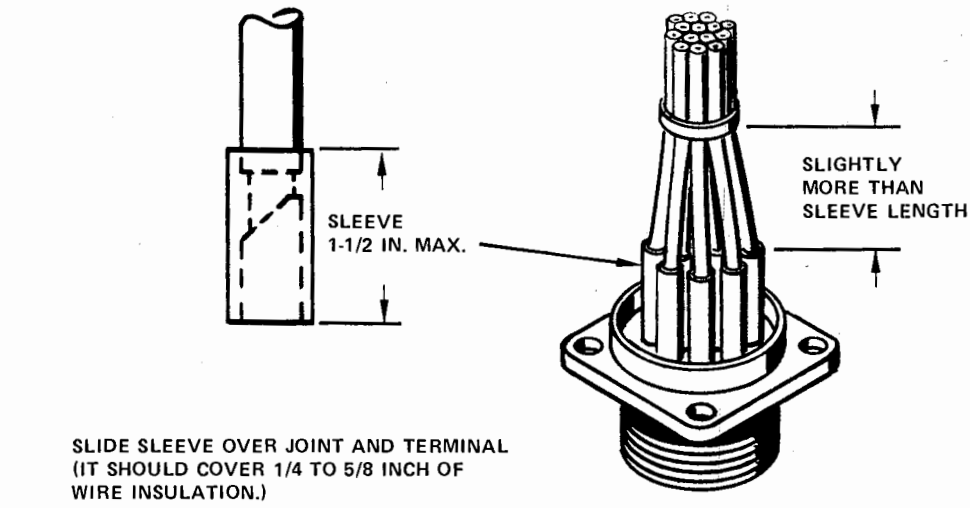
MINIMUM RADIUS BENDS



LEAD BENDING METHOD USING TOOL (PARA 4.4.4)

FIGURE 9. LEAD BENDING (PARA 4.5.2.5)

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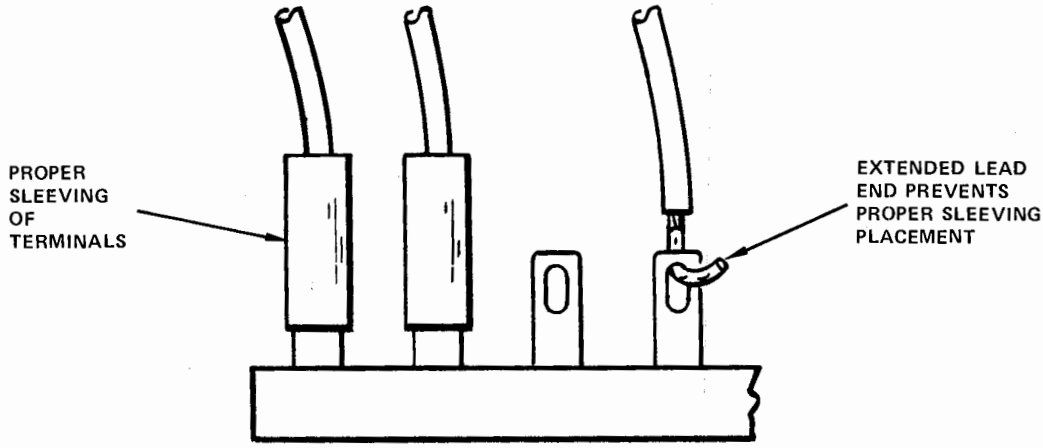
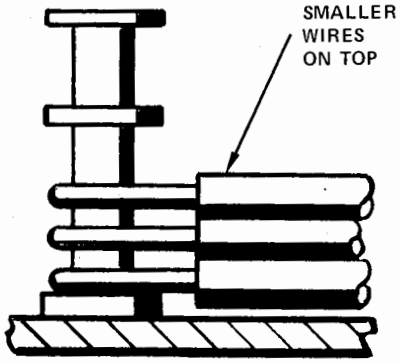


FIGURE 10. SLEEVING APPLICATION (PARA 4.5.4.1)

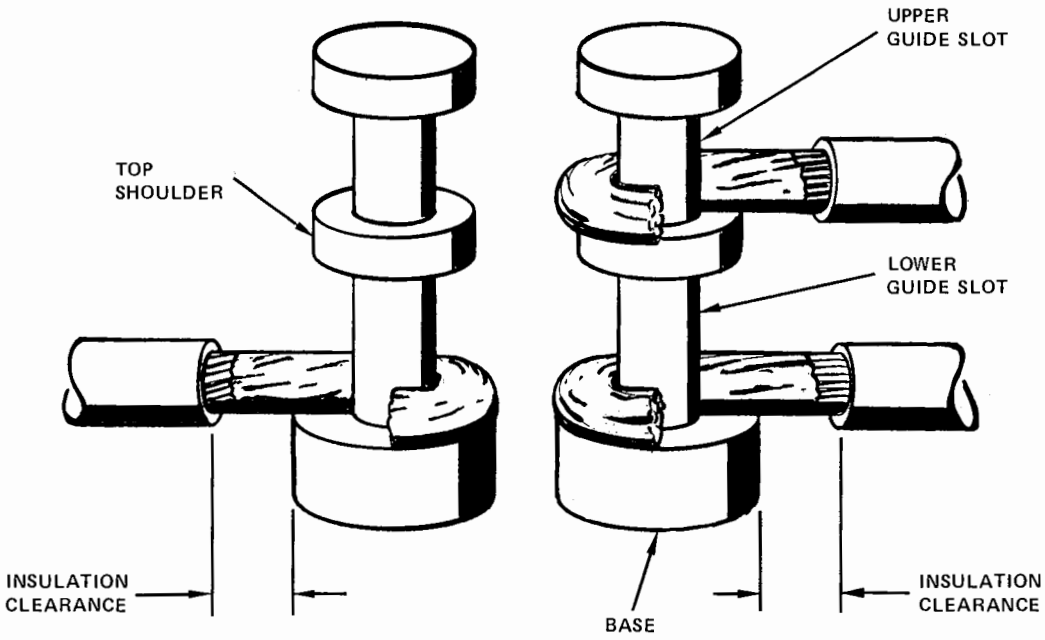
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PARALLEL WIRES
IN MULTIPLE CONNECTION



WRAP ANGLE LIMITS



TYPICAL TURRET TERMINAL USE

FIGURE 11. TURRET TERMINAL WIRE WRAP (PARA 4.5.4.4)

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DESIRED

1. JOINT IS UNDER NO TENSION.
2. ORIGINAL LAY OR TWIST OF THE WIRE HAS BEEN MAINTAINED.
3. LEAD WRAP IS APPROXIMATELY 180°.



ACCEPTABLE

1. DIAMETER OF LOOP IS SUFFICIENT TO AVOID TENSION ON JOINT.
2. ORIGINAL LAY OR TWIST OF THE WIRE HAS BEEN MAINTAINED.
3. LEAD WRAP IS APPROXIMATELY 270°.

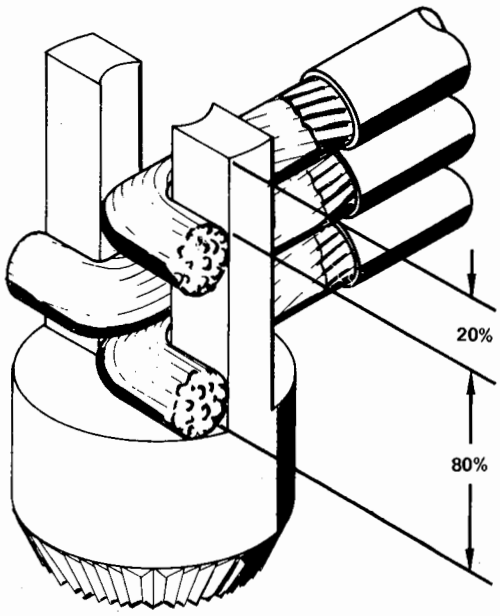


REJECT

1. WIRE HAS BEEN BENT IN A SHARP ANGLE BEND CAUSING STRANDS TO KINK.
2. STRANDS HAVE BEEN FLARED (BIRD-CAGED) ON TERMINAL.
3. INSULATION HAS BEEN CUT.
4. EXCESS PIGTAIL OVERHANG.
5. CONDUCTOR STRANDS HAVE BEEN NICKED.

FIGURE 12. TURRET TERMINAL WIRE WRAP (PARA 4.5.4.4)

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SHOWING MAXIMUM FILL



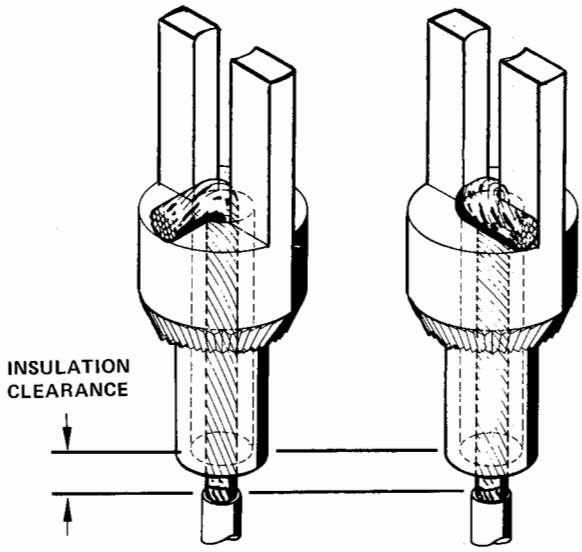
90° MIN

180° MAX

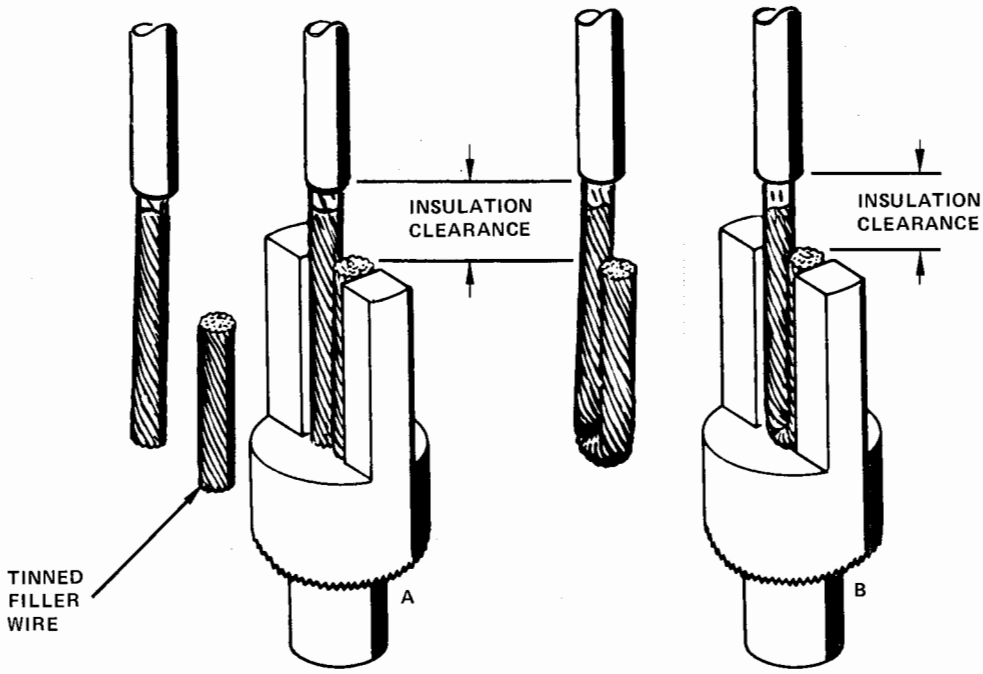
WIRE WRAP LIMITS (DEPENDING ON DESIGN CONFIGURATION AND CLEARANCE WITH ADJACENT PARTS)

FIGURE 13. BIFURCATED TERMINAL WIRE WRAP (PARA 4.5.4.5.1)

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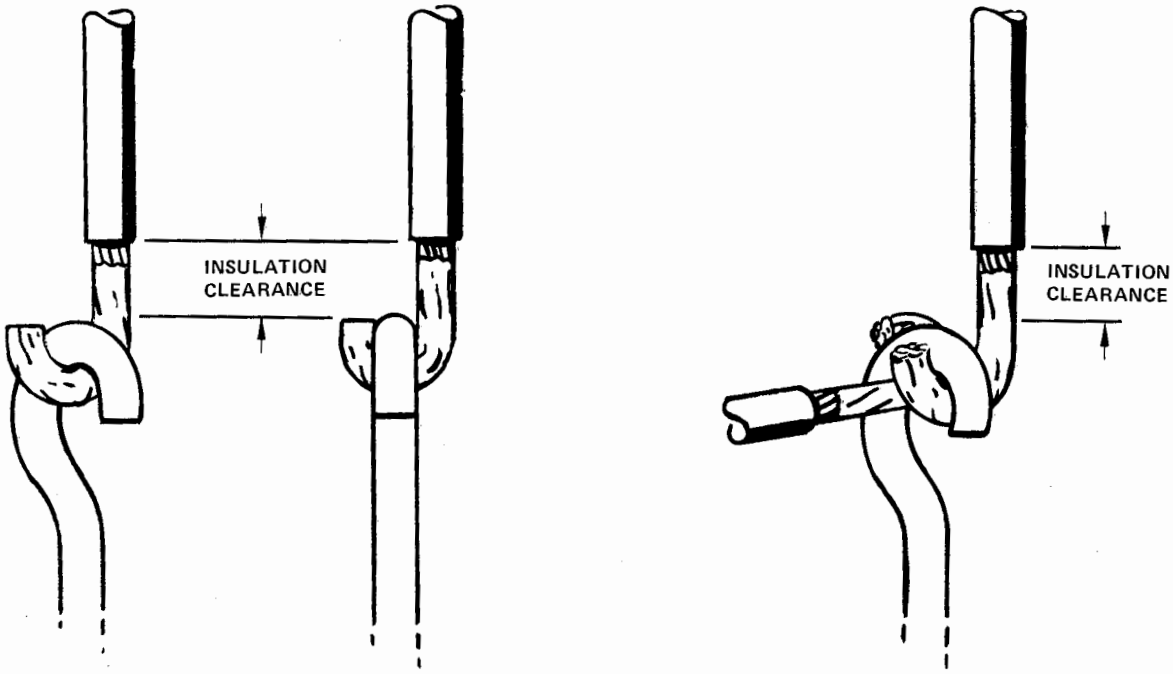
BOTTOM ROUTE CONNECTION



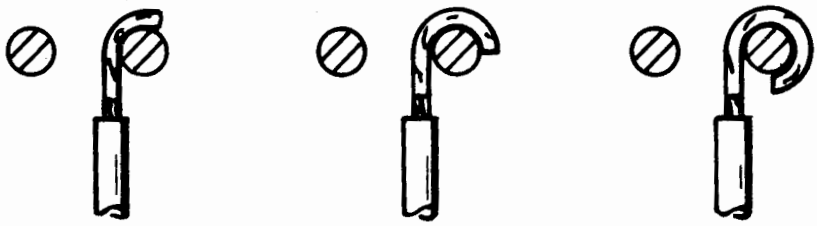
TOP ROUTE CONNECTION

FIGURE 14. BIFURCATED TERMINAL WIRE WRAP (PARA 4.5.4.5.2)

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CONNECTIONS ON HOOK TERMINALS



WIRE WRAP LIMITS

FIGURE 15. HOOK TERMINAL WIRE WRAP (PARA 4.5.4.6)

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ACCEPTABLE TERMINATIONS



1. ORIGINAL LAY OR TWIST OF WIRE HAS BEEN MAINTAINED.
2. LEAD WRAP IS APPROXIMATELY 180°.



1. DOUBLE LEADS ARE DRESSED TO THE TERMINAL FROM OPPOSITE SIDES.

FIGURE 16. HOOK TERMINAL WIRE WRAP (PARA 4.5.4.6)

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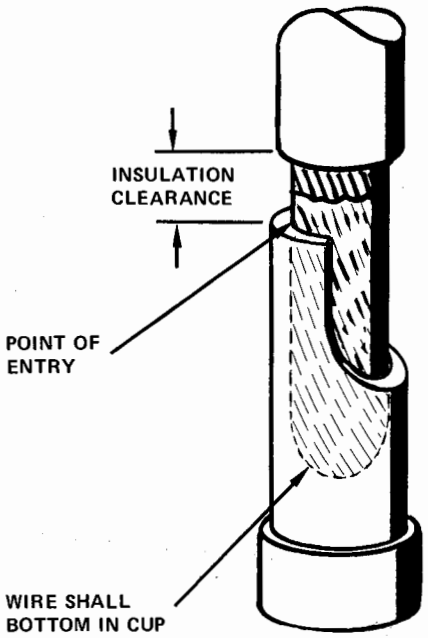


FIGURE 17. SOLDER CUP TERMINAL LEAD PLACEMENT (PARA 4.5.4.7)

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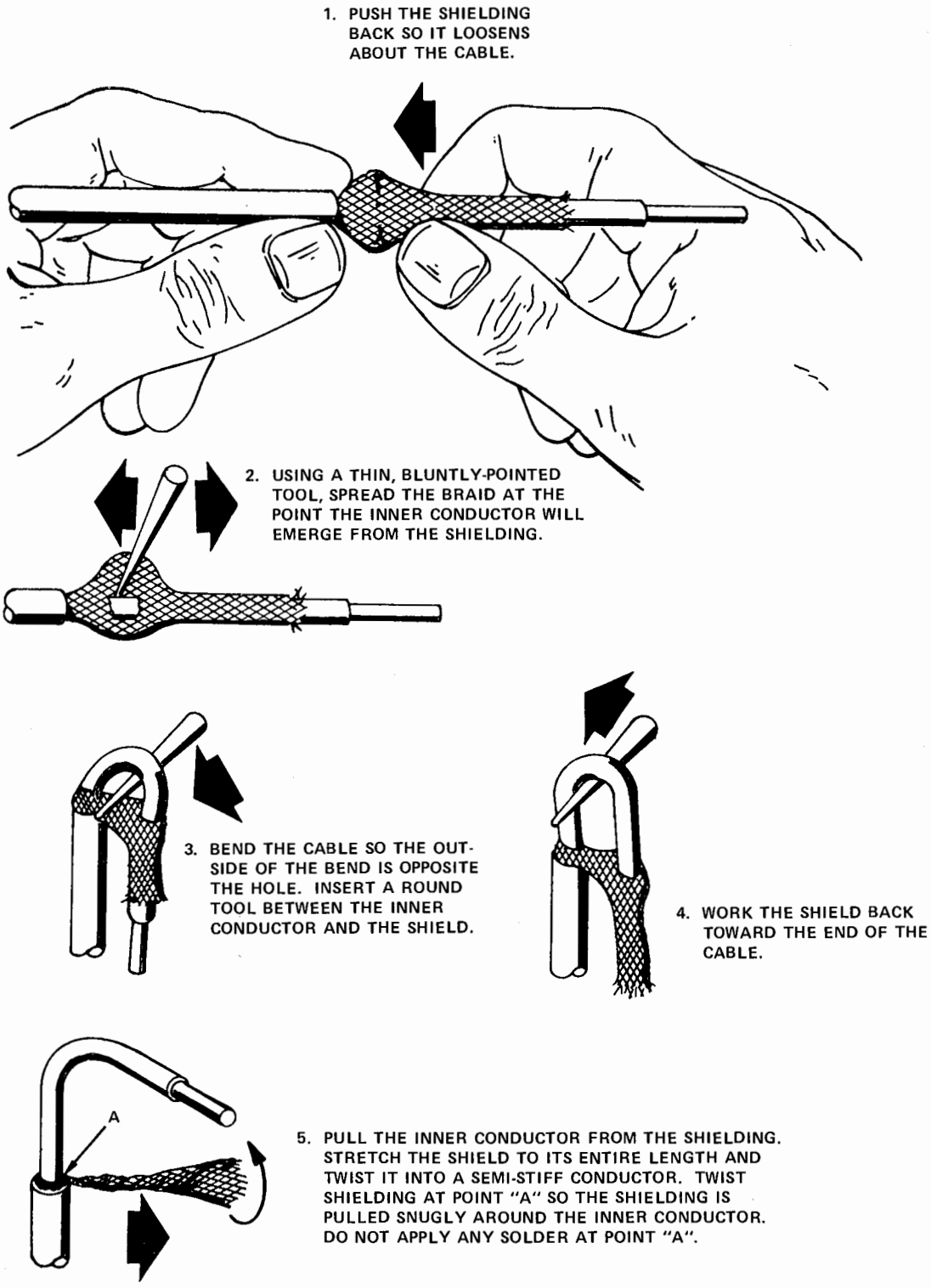
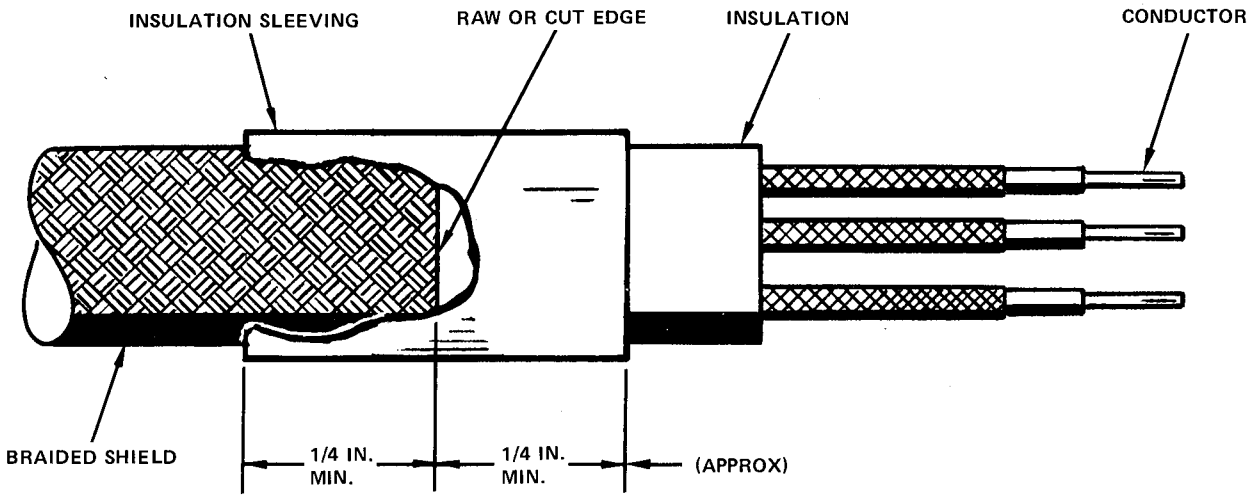
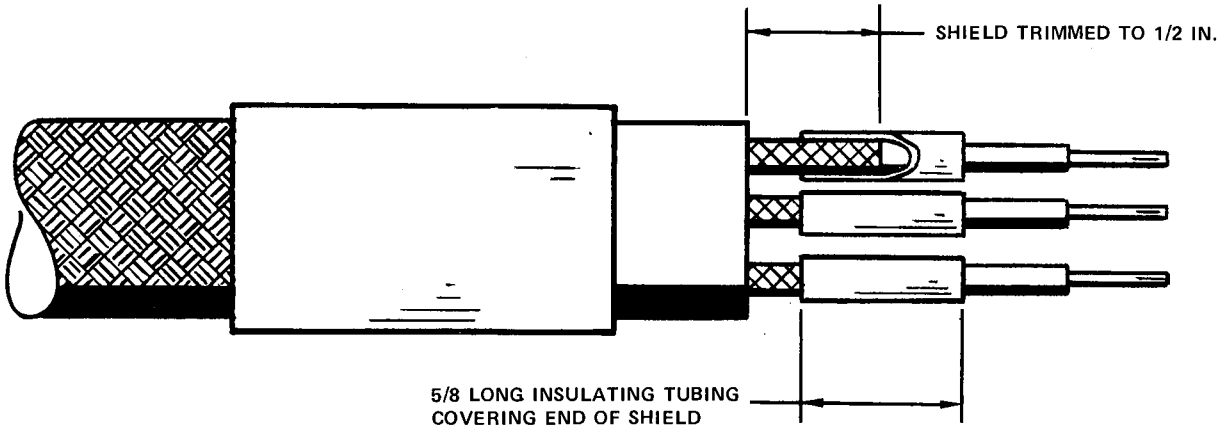


FIGURE 18. SHIELD REMOVAL FOR GROUND TERMINATION (PARA 4.5.5.2)

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TERMINATION OF UNGROUNDED SHIELDING (PARA 4.5.5.3)

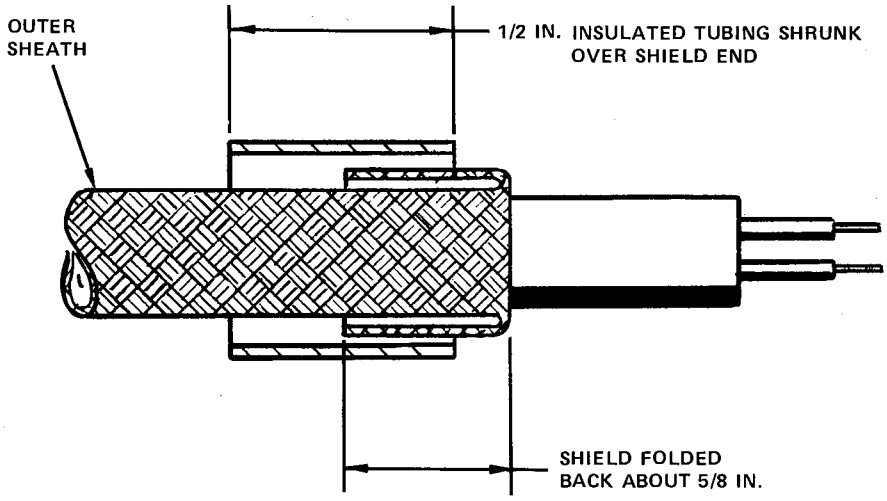


TERMINATION OF SHIELDED SINGLE CONDUCTOR IN MULTI-CONDUCTOR CABLE (PARA 4.5.5.4)

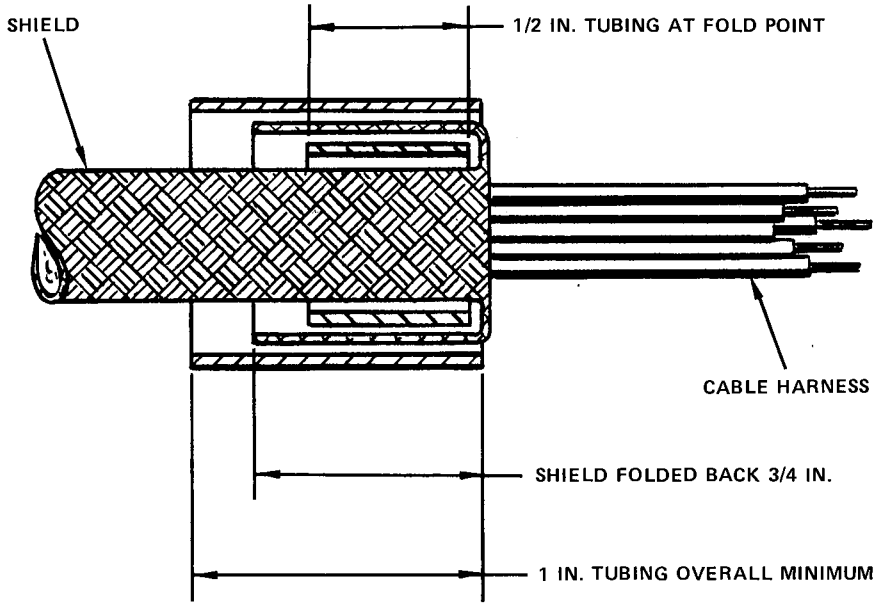
NOTE: DIMENSIONS ON THIS PAGE ARE APPROXIMATE AND WILL VARY WITH CABLE DIAMETER.

FIGURE 19. UNGROUNDED SHIELD TERMINATION

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TERMINATION OF OUTER SHEATHED CABLES (PARA 4.5.5.5)



TERMINATION OF SHIELDS ON CABLE HARNESS (PARA 4.5.5.6)

NOTE: DIMENSIONS ON THIS PAGE ARE APPROXIMATE AND WILL VARY WITH CABLE DIAMETER.

FIGURE 20. UNGROUNDED SHIELD TERMINATION

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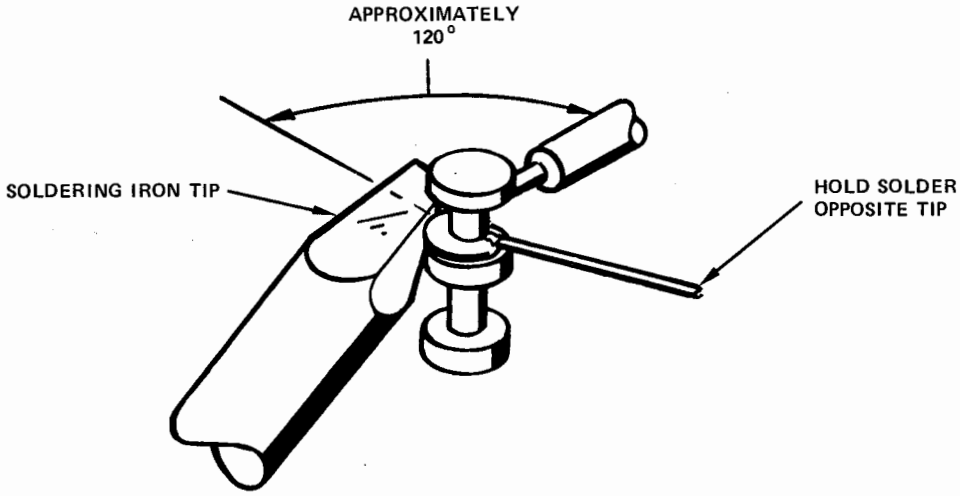


Figure 21A. Positioning Soldering Iron and Applying Solder (para 4.6.2)

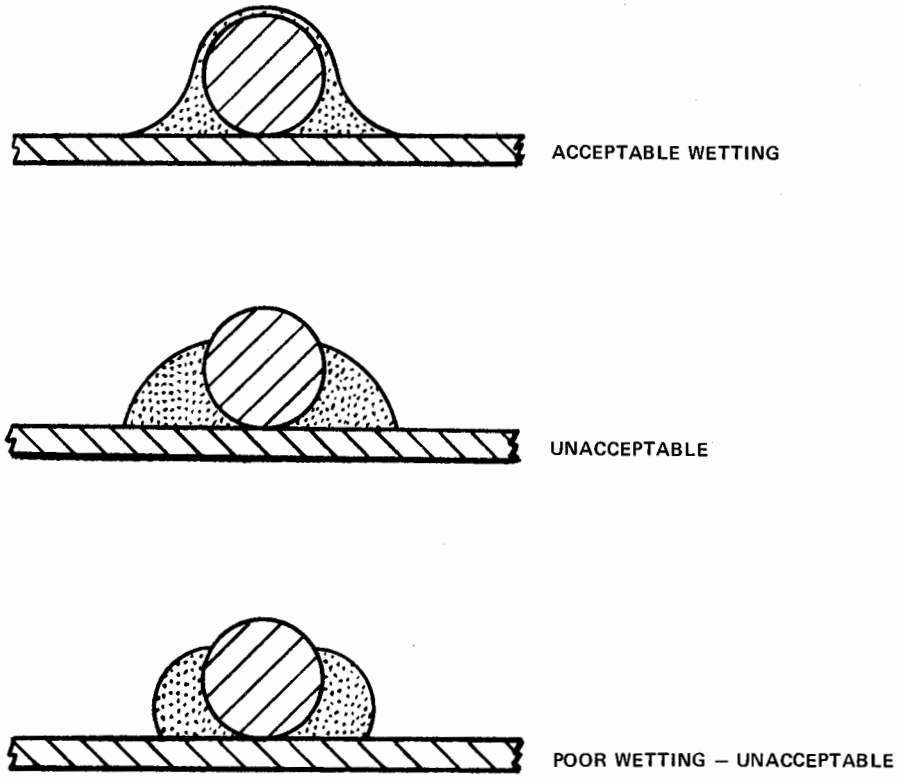


Figure 21B. Wetting Examples (para 4.6.3)

FIGURE 21. SOLDER APPLICATION

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Check the source to verify that this is the current version before use.

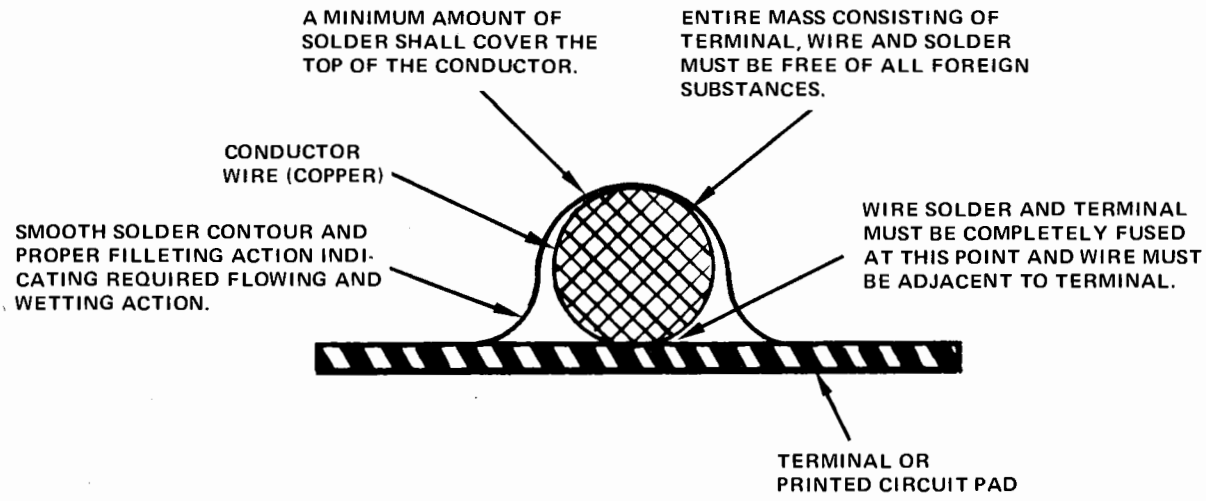


Figure 22A. Printed Circuit Soldering (para 4.6.3.3)

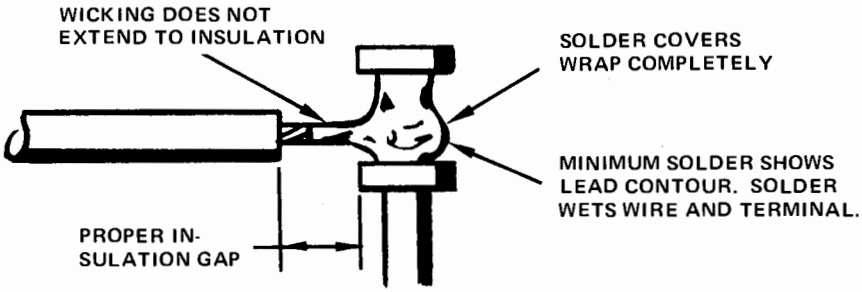


Figure 22B. Turret Soldering (para 4.6.3.1)

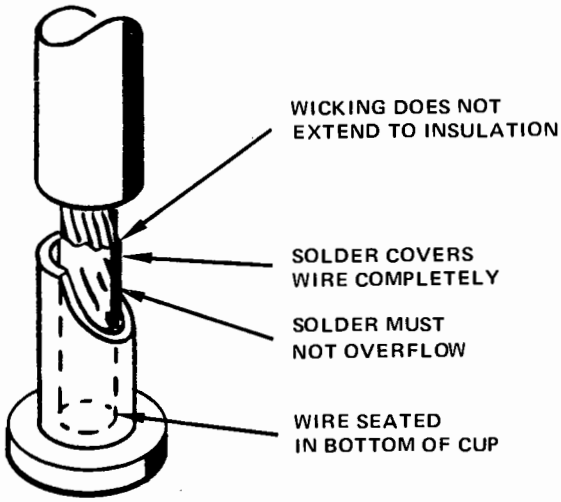
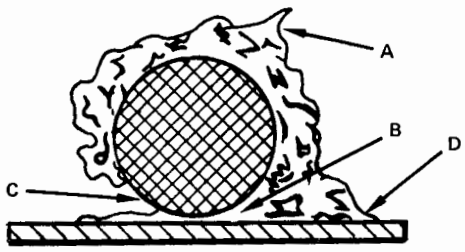


Figure 22C. Solder Cup Soldering (para 4.6.3.2)

FIGURE 22. ACCEPTABLE JOINTS

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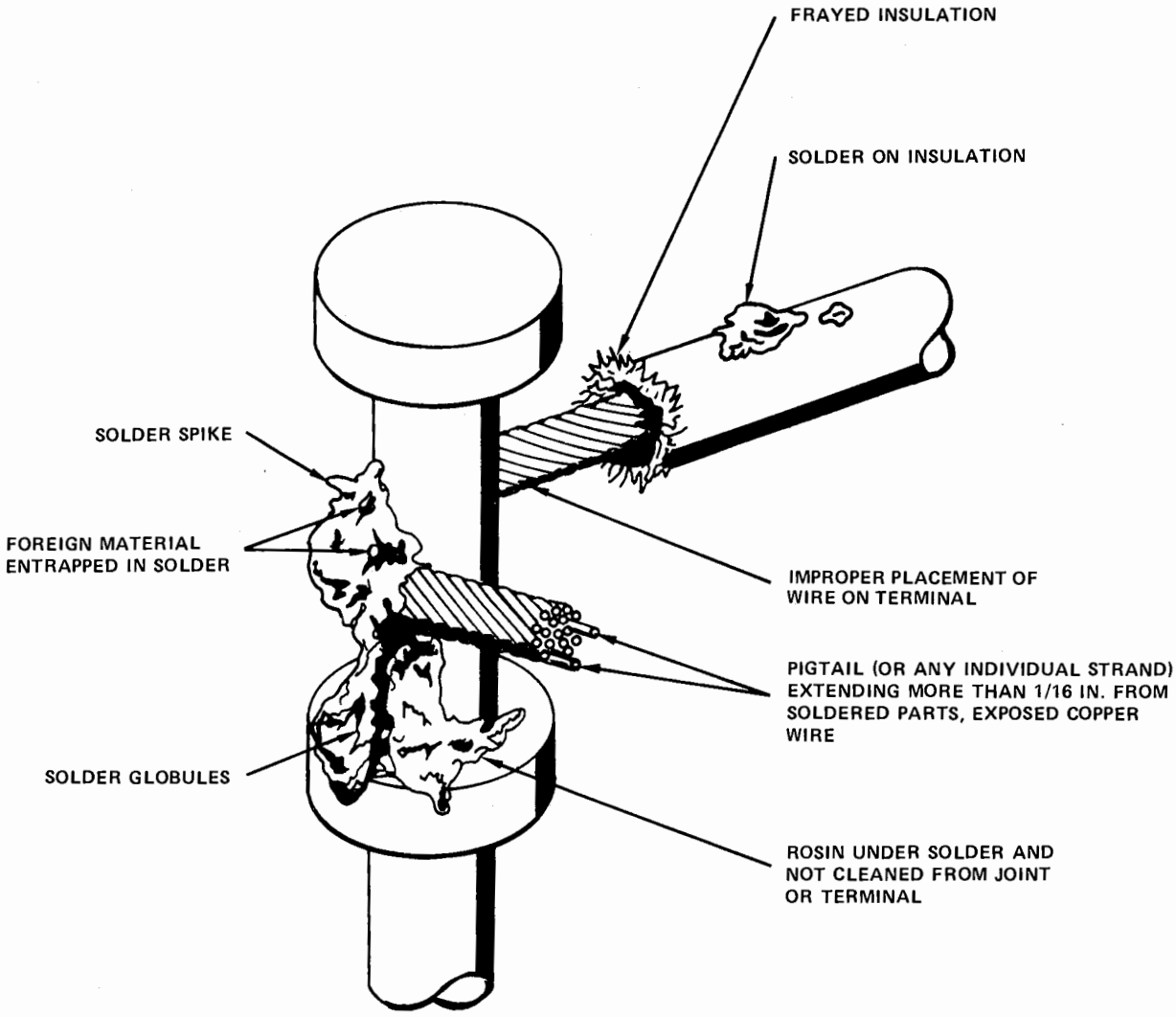
A. SOLDER HAS A CHALKY APPEARANCE, LACKS METALLIC LUSTER, GENERALLY PRESENTS A ROUGH, PILED-UP APPEARANCE.

B. SOLDER HAS NOT BONDED TERMINAL AND WIRE TOGETHER.

C. SOLDER COVERAGE NOT COMPLETE.

D. IMPROPER FILLETING INDICATING INSUFFICIENT FLOWING AND WETTING ACTION.

COLD OR UNDER-HEATED SOLDER JOINT



POOR WORKMANSHIP – ANY ONE OR MORE OF THE CONDITIONS SHOWN CONSTITUTES POOR WORKMANSHIP AND IS UNACCEPTABLE.

FIGURE 23. UNACCEPTABLE JOINTS (PARA 4.6.4)

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DESIRED

1. SOLDER IS SUFFICIENT TO COVER LEAD AND "FEATHER" OUT ON TERMINAL POST.
2. LEAD IS CLEAN WITH NO SURFACE MARKS.



ACCEPTABLE

1. SMALL, SMOOTH SURFACE MARKS IN LEAD. HOWEVER, LEAD IS CLEAN AND WELL TINNED.
2. SOLDER IS MAXIMUM BUT LEAD IS WELL DEFINED. FLOW IS GOOD, TERMINAL AND LEAD ARE WELL WETTED.
3. MINOR IMPERFECTIONS IN SOLDER SURFACE BUT CONNECTION MAINTAINS A BRIGHT APPEARANCE.



REJECT

1. EXCESS SOLDER, LEAD NOT DEFINED.
2. SOLDER HAS A ROUGH FROSTED APPEARANCE, EVIDENCE OF COLD SOLDER JOINT.

FIGURE 24. COMPONENT SOLDERING EXAMPLE (PARA 4.6.3.1)

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DESIRED

1. COMPONENT LEAD IS CLEAN AND WELL TINNED.
2. SOLDER IS NEAT AND BRIGHT AND COMPLETELY COVERS THE WIRE.



ACCEPTABLE

1. DIAMETER OF LEAD WRAP IS SLIGHTLY LARGE BUT LEAD IS WELL BONDED TO TERMINAL.
2. SOLDER IS MAXIMUM BUT LEAD IS WELL DEFINED.



REJECT

1. SOLDER HAS A ROUGH DULL APPEARANCE.
2. SOLDERED PORTION OF LEAD CANNOT BE DISTINGUISHED.
3. COMPONENT LEAD SHOWS EVIDENCE OF CONTAMINATION – SOLDER HAS NOT WET LEAD.
4. BENT COMPONENT LEAD SHOWS EVIDENCE OF CARELESS HANDLING.

FIGURE 25. COMPONENT SOLDERING EXAMPLE (PARA 4.6.3.1)

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DESIRED

1. INSULATION IS CLEAN AND UNDAMAGED.
2.. TRIM IS NEAT AND EVEN.



ACCEPTABLE

1. SLIGHT SCORCH MARKS ON INSULATION.
2. SOLDER IS SLIGHTLY DULL BUT SURFACE IS SMOOTH AND UNIFORM. LEAD AND TERMINAL ARE WELL WETTED.
3. MAXIMUM SOLDER BUT CONTOUR OF CONDUCTOR IS VISIBLE.

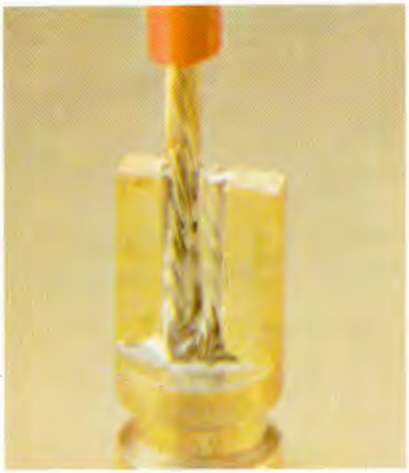


REJECT

1. INSULATION HAS BEEN BURNED.
2. WICKING UNDER INSULATION AS EVIDENCED BY ENLARGED INSULATION.
3. STRANDS HAVE BEEN FLARED ON CONDUCTOR.
4. EXCESS PIGTAIL OVERHANG.

FIGURE 26. INSULATED WIRE SOLDERING EXAMPLE (PARA 4.6.3.1)

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DESIRED

1. INSULATION IS UNMARKED.
2. EXPOSED BARE WIRE IS CLEAN WITH NO EVIDENCE OF OF WICKING.
3. TRIM IS NEAT AND EVEN.



ACCEPTABLE

1. LEAD DOES NOT OVERHANG TERMINAL.
2. EXPOSED WIRE IS WITHIN TOLERANCE WITH NO SIGN OF WICKING.

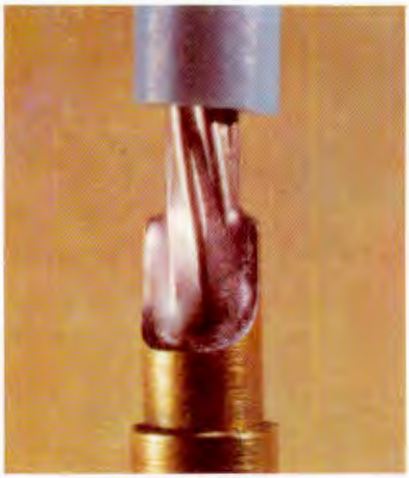


REJECT

1. ORIGINAL LAY OR TWIST OF WIRES HAS NOT BEEN MAINTAINED.
2. INSULATION STRIP IS EXCESSIVE AND IRREGULAR.
3. EXCESS PIGTAIL OVERHANG.
4. INSUFFICIENT SOLDER – LEAD AND TERMINAL ARE NOT WELL WETTED.
5. GENERAL POOR QUALITY THROUGHOUT.

FIGURE 27. INSULATED WIRE SOLDERING EXAMPLE (PARA 4.6.3.1)

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DESIRED

1. CONDUCTOR IS CORRECT SIZE FOR CUP APPLICATION.
2. SOLDER COMPLETELY FILLS THE CUP AND FOLLOWS THE CONTOUR OF THE CUP ENTRY SLOT.
3. STRANDS ARE WELL DEFINED BELOW THE TOP SURFACE OF THE CUP.



ACCEPTABLE

1. SOLDER HAS WICKED UP TO, BUT NOT UNDER THE INSULATION. NO EVIDENCE OF SCORCHING OR ENLARGING OF THE INSULATION.
2. ALL STRANDS ARE WITHIN CUP AND FULLY SEATED.
3. MAXIMUM SOLDER; HOWEVER THE SOLDER DOES NOT EXTEND BEYOND CUP DIAMETER.
4. STRANDS ARE DEFINED BELOW THE TOP SURFACE OF THE CUP.



REJECT

1. CONDUCTOR IS TOO BIG FOR CUP APPLICATION.
2. CONNECTION CONTAINS LOOSE STRANDS AND HAVE BEEN CUT TO REDUCE SIZE OF CONDUCTOR TO FIT CUP.
3. INSULATION STRIP IS EXCESSIVE.

FIGURE 28. CONNECTOR CUP SOLDERING EXAMPLES (PARA 4.6.3.2)

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DESIRED

1. ONLY SUFFICIENT SOLDER HAS BEEN USED TO FILL THE CUP.
2. CONDUCTOR SHOWS NO SIGN OF WICKING, INDIVIDUAL STRANDS ARE VISIBLE.



ACCEPTABLE

1. CUP HAS BEEN "SPOT TINNED" AS A RESULT OF THE SOLDERING IRON APPLICATION.
2. MAXIMUM AMOUNT OF SOLDER HAS BEEN USED BUT DOES NOT PROJECT BEYOND CUP DIAMETER.
3. NO EVIDENCE OF "WICKING".



REJECT

1. COLD SOLDER.
2. EXCESS SOLDER OBSCURES STRANDS OF WIRE IN THE CUP.
3. WICKING IS EVIDENT BY SOLDER EXTENDING TO AND UNDER INSULATION AS EVIDENCED BY ENLARGED INSULATION.

FIGURE 29. CONNECTOR CUP SOLDERING EXAMPLES (PARA 4.6.3.2)

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DESIRED

1. LEAD AND PAD ARE WELL WETTED.
2. CONTOUR OF LEAD IS CLEARLY DEFINED.
3. SOLDER HAS A SMOOTH, SHINY APPEARANCE.



ACCEPTABLE

1. SMOOTH, METALLIC APPEARANCE.
2. MAXIMUM SOLDER BUT LEAD IS VISIBLE.



REJECT

1. EVIDENCE OF CONTAMINATION. LEAD IS NOT SOLDERED.
2. EXCESS SOLDER EXTENDING BEYOND PAD DIAMETER.
3. PAD HAS NOT BEEN PROPERLY WETTED UNDER LEAD.

FIGURE 30. PRINTED CIRCUIT SOLDERING EXAMPLES (PARA 4.6.3.3)

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DESIRED

1. NO SURFACE IMPERFECTIONS.
2. CONCAVE SHAPE, BRIGHT SOLDER.
3. GOOD WETTING OF PIN AND PAD.



ACCEPTABLE

1. MINOR SURFACE IMPERFECTIONS.
2. SLIGHTLY DULL APPEARANCE. HOWEVER, SOLDER FLOW IS NOT BROKEN.
3. MAXIMUM SOLDER BUT PIN CONTOUR IS VISIBLE.
4. PIN IS COMPLETELY TINNED (PIN DOES NOT HAVE TO BE COMPLETELY TINNED AT END OF TIP WHEN UNIT IS TO BE CONFORMAL COATED. COATING PRECLUDES THE POSSIBILITY OF OXIDATION).



REJECT

1. EVIDENCE OF IMPROPER WETTING, SOLDER DOES NOT COVER PAD, BASE METAL SHOWING.
2. SOLDER INCOMPLETE ON TIP OF PIN. BASE METAL SHOWING. BREAK BETWEEN PIN AND SOLDER FILLET.

FIGURE 31. PRINTED CIRCUIT SOLDERING EXAMPLES (PARA 4.6.3.3)

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Check the source to verify that this is the current version before use.

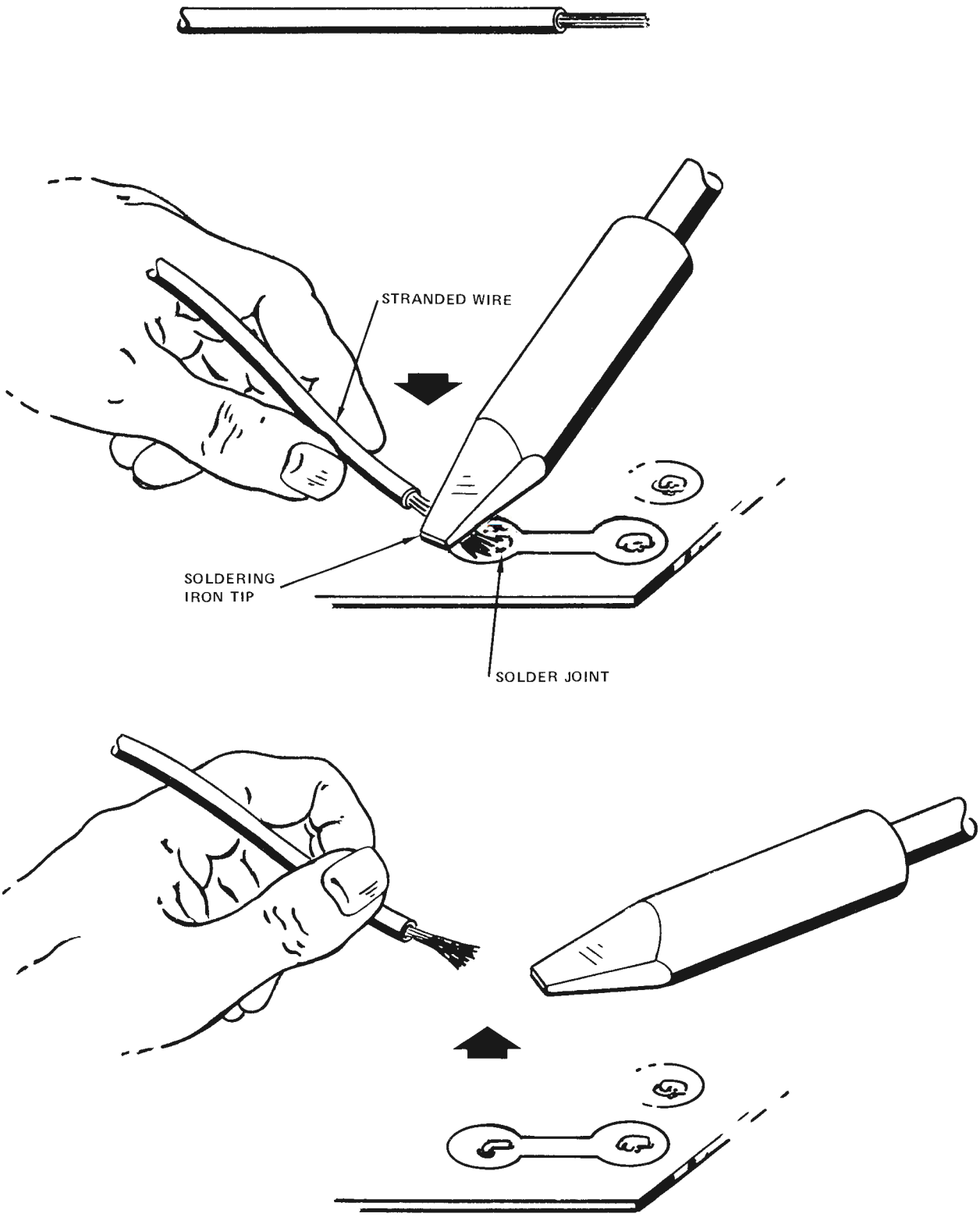


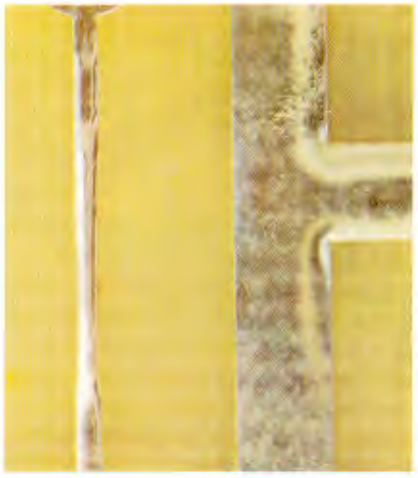
FIGURE 32. USING WICK TO REMOVE SOLDER (PARA 4.6.7.2)

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Check the source to verify that this is the current version before use.



DESIRED

1. SOLDER HAS TINNED THE CONDUCTOR THINLY AND EVENLY.
2. SURFACE HAS A SMOOTH SHINY APPEARANCE.



ACCEPTABLE

1. TINNING IS NOT UNIFORM IN THICKNESS BUT ADEQUATE PROTECTION IS PROVIDED FOR CIRCUIT PATH.

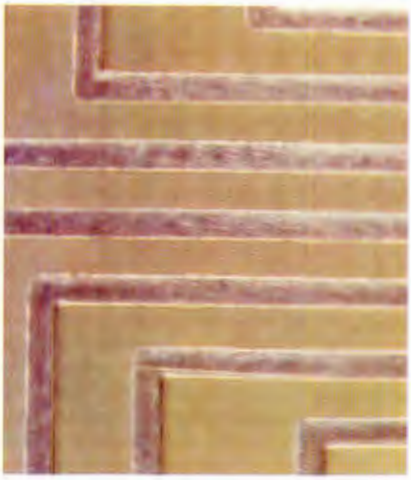


REJECT

1. CIRCUIT PATHS ARE IMPROPERLY WETTED EXPOSING BASE METAL.

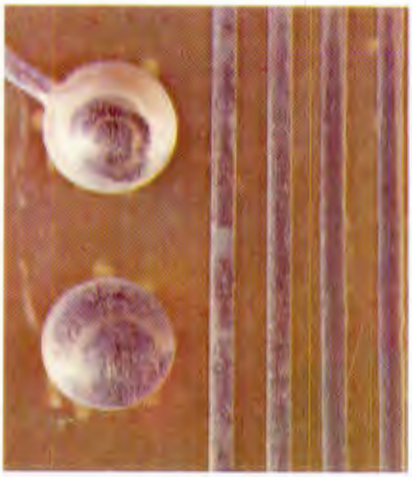
FIGURE 33. PRINTED CIRCUIT PATTERN TINNING (PARA 4.7.3)

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DESIRED

1. CONDUCTOR IS CLEAN AND SURFACE IS UNMARKED.
2. SOLDER HAS TINNED THE CONDUCTOR THINLY AND EVENLY.



ACCEPTABLE

1. CIRCUIT PATHS AND PADS HAVE MAXIMUM SOLDER COATING. SOLDER COATING DOES NOT EXTEND BEYOND PAD DIAMETERS.

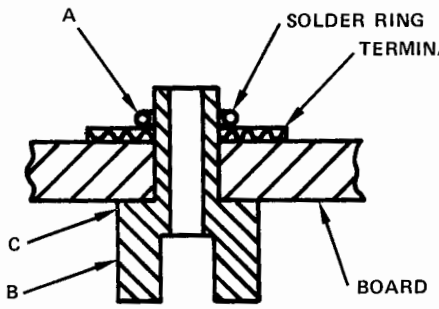


REJECT

1. SOLDER IS THICK AND UNEVEN.
2. ICICLES EXTEND ABOVE AND OUT BEYOND THE CIRCUIT PATH SURFACES. POTENTIAL SHORT.

FIGURE 34. PRINTED CIRCUIT PATTERN TINNING (PARA 4.7.3)

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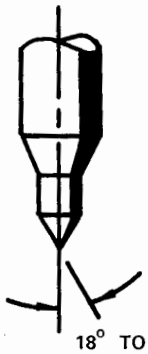


A. SOLDER RING WITH FLUX IS USED.

B. TERMINAL READY FOR SWAGING.

C. TERMINAL MOUNTING SHOULDER.

TERMINAL BEFORE SWAGING (PARA 4.7.5.2)

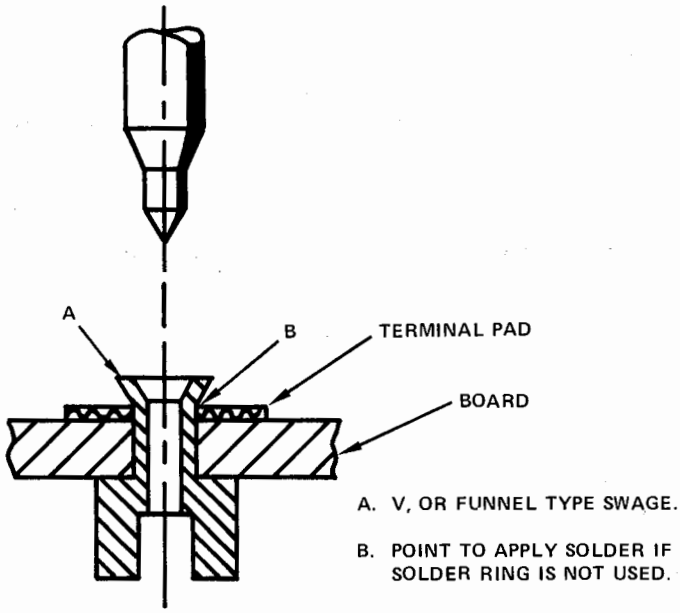


18° TO 30°, DEPENDING ON THICKNESS, HARDNESS AND LENGTH OF TERMINAL MOUNTING SHANK.

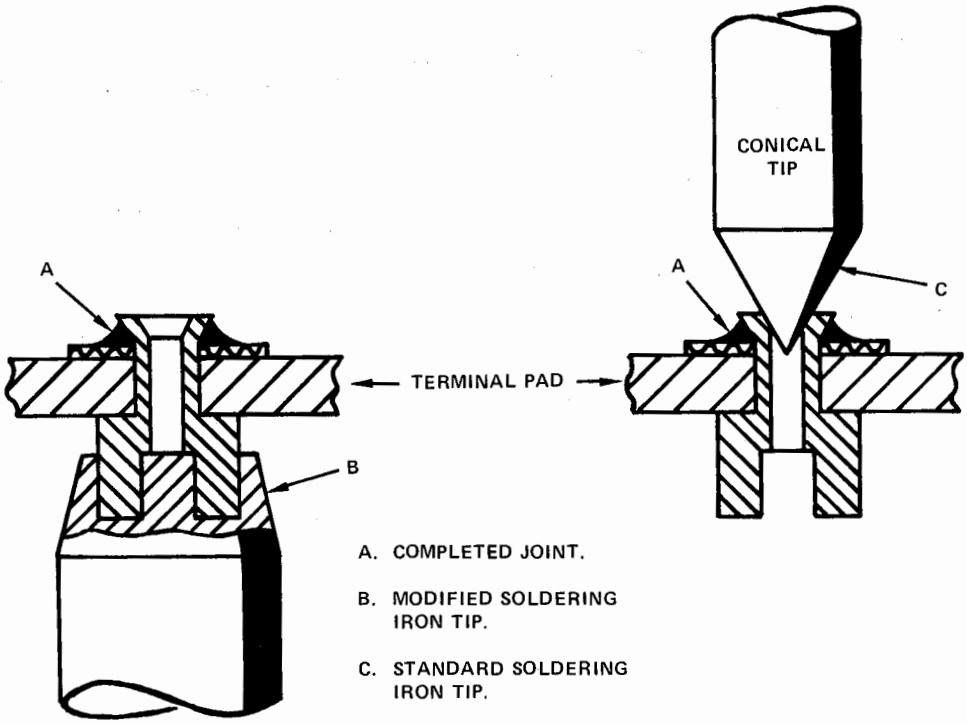
TYPICAL V-TYPE SWAGING TOOL (PARA 4.7.5.3.2)

FIGURE 35. TERMINAL SWAGING PREPARATION

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TYPICAL SWAGED TERMINAL (PARA 4.7.5.3.2)



SOLDERING SWAGED TERMINALS (PARA 4.7.5.4.1)

FIGURE 36. SWAGED TERMINALS

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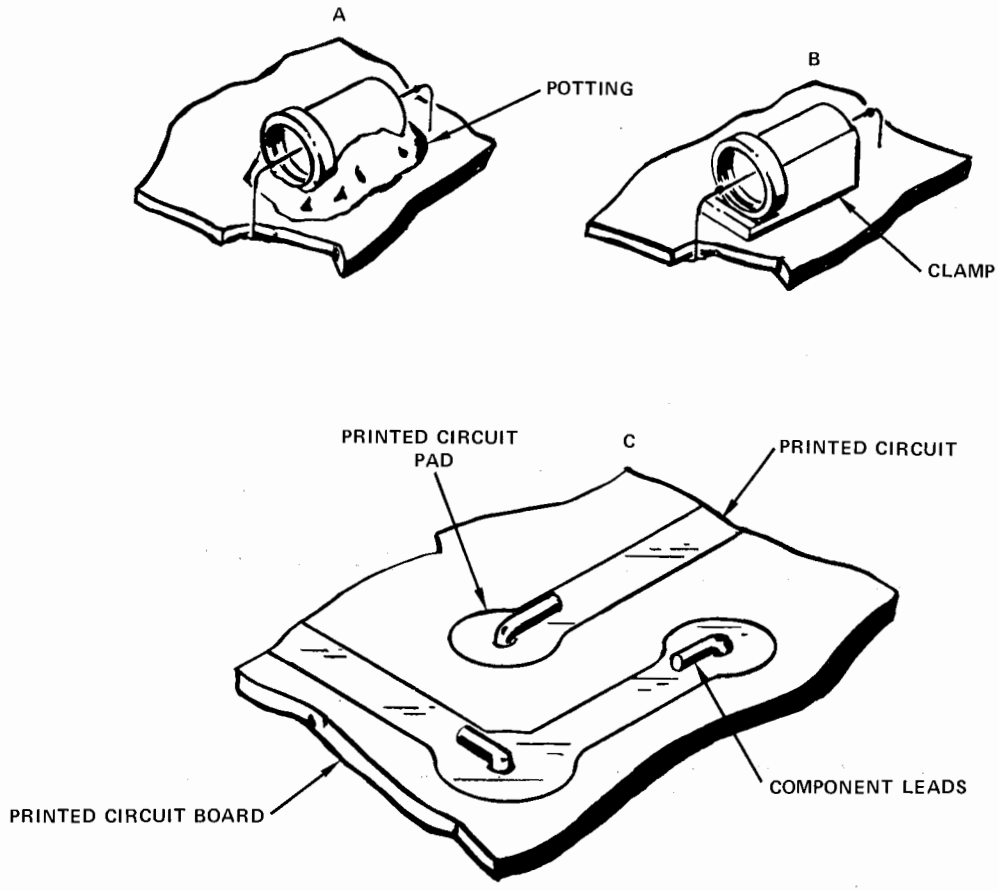


FIGURE 37. EXAMPLES OF COMPONENT MOUNTING (PARA 4.7.6.3)

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DESIRED

1. LEAD IS CLINCHED IN LINE WITH THE CIRCUIT PATH.
2. LEAD IS CLINCHED FLUSH TO CIRCUIT.
3. LEAD IS APPROXIMATELY ONE RADIUS LONG.



ACCEPTABLE

1. LEAD IS POINTING TOWARD CIRCUIT PATH.
2. LEAD IS CLINCHED FLUSH TO CIRCUIT WITHIN ONE LEAD DIAMETER.



REJECT

1. LEAD EXTENDS BEYOND EDGE OF CIRCUIT PATTERN.

FIGURE 38. CLINCHED LEAD EXAMPLES (PARA 4.7.6.3)

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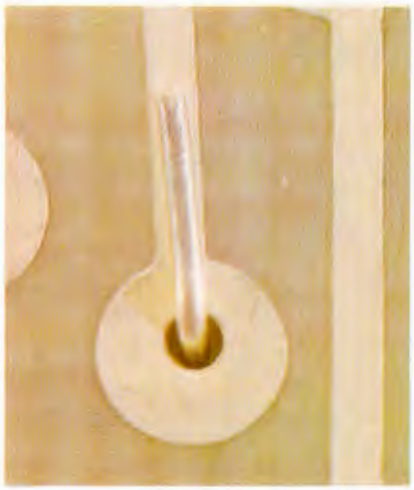
DESIRED

1. LEAD CLINCHED IN LINE WITH CIRCUIT PATH.
2. BENT OVER PORTION OF LEAD IS NOT MORE THAN ONE PAD DIAMETER LONG.
3. LEAD IS CLINCHED FLUSH TO CIRCUIT.



ACCEPTABLE

1. LEAD IS APPROXIMATELY ONE RADIUS LONG.
2. LEAD IS CLINCHED FLUSH TO CIRCUIT.

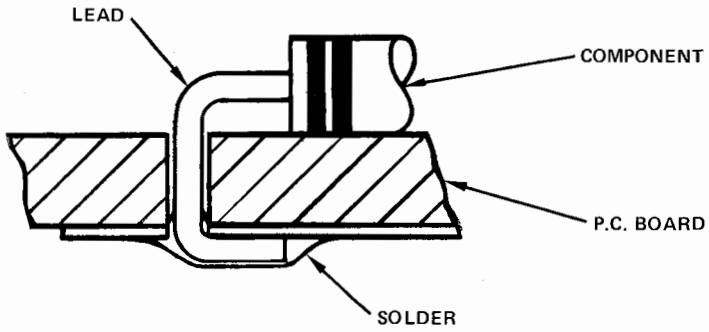


REJECT

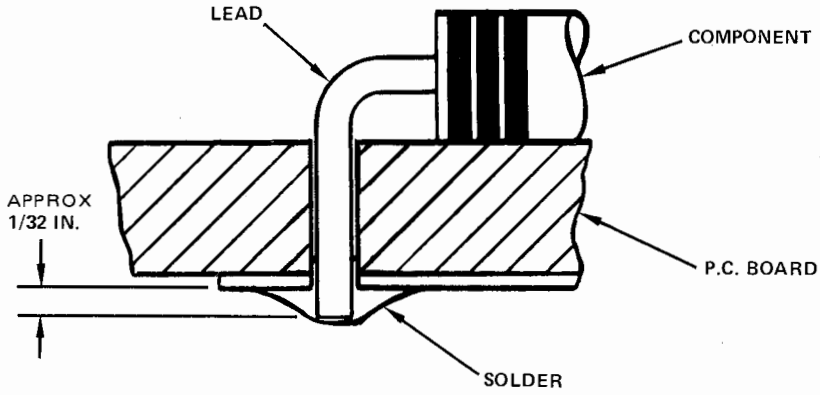
1. LEAD IS MORE THAN ONE PAD DIAMETER LONG.

FIGURE 39. CLINCHED LEAD EXAMPLES (PARA 4.7.6.3)

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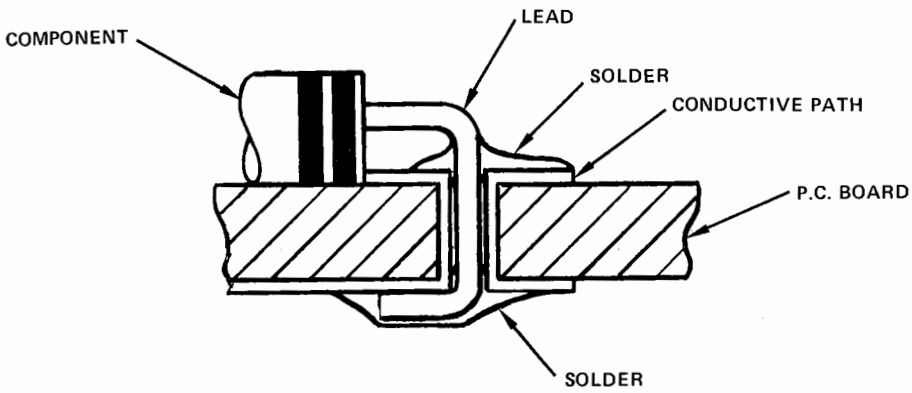
A. STANDARD (PARA 4.7.6.3)



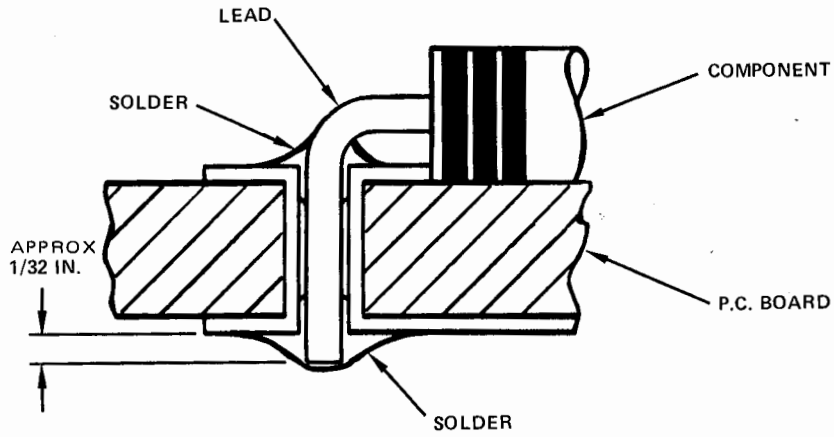
B. SPECIAL (PARA 4.7.6.4)

FIGURE 40. COMPONENT LEAD TERMINATION, CIRCUIT ON ONE SIDE

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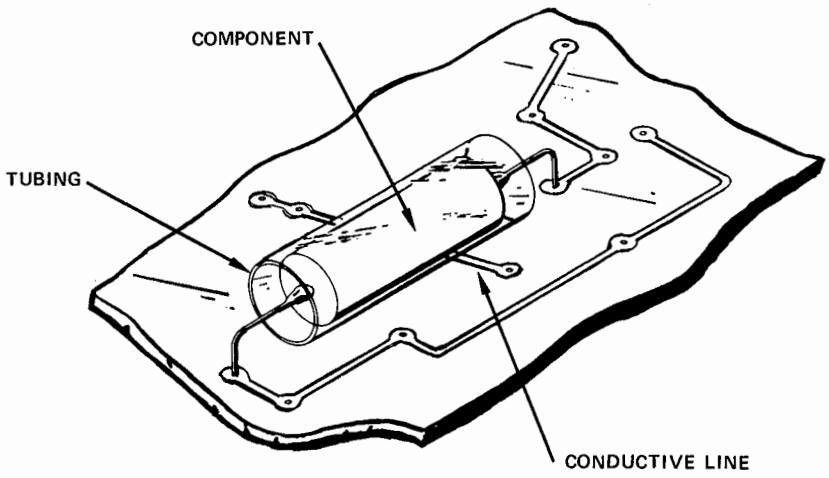
A. STANDARD



B. SPECIAL

FIGURE 41. COMPONENT LEAD TERMINATION, CIRCUIT ON BOTH SIDES (PARA 4.7.6.5)

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CROSSING CONDUCTIVE LINES (PARA 4.7.6.1.2)

FIGURE 42. CROSSING CONDUCTIVE LINES (PARA 4.7.6.1.2)

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DESIRED

1. LEAD AND PAD ARE WELL WETTED.
2. CONTOUR OF LEAD IS CLEARLY DEFINED.
3. SOLDER HAS A SMOOTH, SHINY APPEARANCE.



ACCEPTABLE

1. SMALL IMPERFECTIONS IN SURFACE.
2. SMOOTH METALLIC APPEARANCE.
3. MAXIMUM SOLDER BUT LEAD IS VISIBLE.



REJECT

1. EVIDENCE OF CONTAMINATION, LEAD IS NOT SOLDERED.
2. PIN HOLE ADJACENT TO LEAD.
3. SOLDER HAS A ROUGH FROSTED APPEARANCE – EVIDENCE OF COLD SOLDER CONNECTION.

FIGURE 43. CLINCHED LEAD SOLDER EXAMPLE (PARA 4.7.8)

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DESIRED

1. NO VOID AREAS OR SURFACE IMPERFECTIONS.
2. 100% SOLDER FILLET AROUND LEAD.
3. SOLDER COVERS LEAD AND FEATHERS OUT TO A THIN EDGE ON CIRCUIT PATH.



ACCEPTABLE

1. MAXIMUM SOLDER.
2. FULL SOLDER FILLET BETWEEN LEAD AND CIRCUIT PATH AT ACCESS HOLE.
3. CLINCHED PORTION OF LEAD IS WELL WETTED AND SOLDERED TO CIRCUIT FOIL.



REJECT

1. FULL SOLDER DOES NOT EXIST BETWEEN LEAD AND CIRCUIT PAD AT ACCESS HOLE.
2. SOLDER FLOW AND WETTING ACTION IS INADEQUATE.

SOLDER COVERAGE (STRAIGHT THROUGH LEAD)

FIGURE 44. CLINCHED LEAD SOLDER EXAMPLE (PARA 4.7.8)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:47Z
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DESIRED

1. SOLDER FILLET AROUND LEAD IS 100% COMPLETE.
2. SOLDER HAS A SMOOTH, BRIGHT METALLIC APPEARANCE.
3. NO VOID AREAS OR PINHOLES. SOLDER FLOW AND WETTING ACTION IS GOOD.



ACCEPTABLE

1. SMALL SURFACE IMPERFECTIONS AROUND PERIPHERY OF CIRCUIT PAD.
2. MINIMUM LEAD LENGTH ABOVE CIRCUIT PAD SHALL BE .025 WITH A MAXIMUM OF .050.
3. SOLDER SHALL COVER ENTIRE END WITH NO BARE CONDUCTOR SHOWING. (PIN DOES NOT HAVE TO BE COMPLETELY TINNED AT END OR TIP WHEN UNIT IS TO BE CONFORMAL COATED. COATING PRECLUDES THE POSSIBILITY OF OXIDATION.)

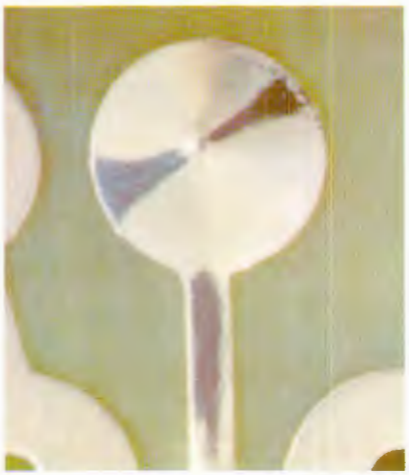


REJECT

1. EXCESS SOLDER, PIN IS NOT WELL DEFINED.
2. COLD SOLDER CONNECTION, SOLDER SURFACE IS ROUGH AND UNEVEN.

FIGURE 45. STRAIGHT LEAD SOLDER EXAMPLE (PARA 4.7.8)

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DESIRED

1. NO SURFACE IMPERFECTIONS.
2. NEAT, BRIGHT SOLDER APPEARANCE.
3. GOOD WETTING OF PIN AND PAD.
4. PIN CONTOUR IS WELL DEFINED.



ACCEPTABLE

1. MINOR SURFACE IMPERFECTIONS.
2. VISIBLE LINE OF DEMARCATION BETWEEN SOLDER AND PIN. HOWEVER, SOLDER FLOW IS NOT BROKEN. PIN IS COMPLETELY TINNED.
3. MINOR VOIDS DO NOT REDUCE SOLDER FILLET AROUND PIN BY MORE THAN 20%.



REJECT

1. EVIDENCE OF IMPROPER WETTING.
2. SOLDER DOES NOT COVER PAD AND PIN. BASE METAL SHOWING.

FIGURE 46. STRAIGHT LEAD SOLDER EXAMPLE (PARA 4.7.8)

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